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Linear Additive Models:
Calibration, Monte Carlo, Exotics and Risk Management

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Introduction

This report studies the calibration, simulation, pricing and risk management of exotic derivatives under three linear additive models: the Minimal Additive model, the Generalized Logistic model and the Additive Bachelier model. The analysis is based on WTI crude oil option market data observed on 2nd June 2020, during the highly volatile period following the Covid-19-related stress in the oil market. This market environment is particularly suitable for a Bachelier-type framework, since oil futures and options in that period required models able to handle very low, and potentially negative, forward prices.

The dataset consists of European call and put options mid prices quoted on a grid of strikes for several maturities. For each expiry, call and put prices are available separately, while the corresponding futures expiry dates are used to construct the time-to-maturity structure. Starting from these market quotes, we first extract the model-independent inputs of the project: market-implied discount factors, forward prices, zero rates and deterministic absolute dividends.

We work in a linear additive setting, where the forward price is written as

$$F(t, T) = F(0, T) + f_t,$$

with f_t described by an additive process. This framework combines analytical tractability with enough flexibility to reproduce non-Gaussian features of the implied volatility surface, such as skewness and exponential tails. The three models considered in the project share the same additive structure, but differ in the distributional assumptions used for the normalized driving factor.

The calibration follows a cascade logic: first the market-implied forwards and discount factors are fixed following [2], then the ATM volatility term structure is extracted, and finally the remaining shape parameters are fitted to the OTM volatility surface.

After calibration, the relevant increments between the reset dates $T_1 = 6m$ and $T_2 = 1y$ are simulated. Since the characteristic functions of the additive increments are available in closed form, the simulation is performed through a Lewis-FFT inversion of the CDF, following the approach of [5]. Particular attention is paid to the numerical stability of the CDF reconstruction, to the extrapolation outside the FFT range, and to the probability atom appearing in the Minimal Additive model.

The calibrated models are then used to price the exotic products required in the assignment: a call-on-call, a put-on-put and a chooser option. Since these payoffs naturally lead to a nested Monte Carlo structure, we implement efficient alternatives based on interpolation and on a no-grid method using sorted inner increments. The resulting prices are compared across models in order to assess the impact of the different distributional

assumptions.

Finally, we study the risk management of a short portfolio of exotic options. The hedge is built using plain vanilla options and the front future, with a quantitative rule based on finite-difference Delta and Vega sensitivities. The effectiveness of the hedge is assessed by comparing the unhedged and hedged P&L on the following two Tuesdays, including transaction costs.

Overall, the aim of the project is to evaluate the practical use of linear additive models for the full workflow of exotic derivative pricing: from market calibration to Monte Carlo simulation, from pricing to hedging.

0 Asymptotic Properties of the Models

In the linear additive framework, the forward price is modeled as

$$F(t, T) = F(0, T) + f_t,$$

where $\{f_t\}_{t \geq 0}$ is an additive process with independent increments. Following the notation of the reference model, we separate the scale of the process from its distributional shape by writing

$$f_t = \sigma_t \sqrt{t} \zeta_t.$$

The random variable ζ_t is the normalized factor: its distribution determines the shape of the smile, while the scale $\sigma_t \sqrt{t}$ determines the level of volatility at maturity t .

This decomposition is useful because the asymptotic tail coefficients are first identified for the normalized variable ζ_t , and then transferred to f_t through the scaling factor. If the normalized density has left and right tail coefficients p^- and p^+ , then the corresponding coefficients for f_t are

$$p_t^- = \frac{p^-}{\sigma_t \sqrt{t}}, \quad p_t^+ = \frac{p^+}{\sigma_t \sqrt{t}}.$$

Therefore, the parameters controlling the shape of the distribution determine the relative tail behaviour, while the time-dependent scale $\sigma_t \sqrt{t}$ controls the absolute decay rate of the tails of f_t .

0.1 Exponential Tail Coefficients

The objective of this point is to determine the exponential tail coefficients of f_t for the three models considered in the project. We denote them by p_t^- and p_t^+ , where p_t^- controls the left tail and p_t^+ controls the right tail. More precisely, letting $\rho_{f_t}(x)$ denote the

probability density function of f_t , the asymptotic tail behaviour is:

$$\rho_{f_t}(x) \sim C^- e^{p_t^- x}, \quad x \rightarrow -\infty,$$

and

$$\rho_{f_t}(x) \sim C^+ e^{-p_t^+ x}, \quad x \rightarrow +\infty.$$

Equivalently, these coefficients correspond to the boundaries of the analyticity strip of the characteristic function,

$$\Im(u) \in (-p_t^+, p_t^-).$$

For the Additive Bachelier model, the normalized tail coefficients are

$$p_{AB}^- = -\eta + \sqrt{\eta^2 + 2\frac{1-\alpha}{\kappa}}, \quad p_{AB}^+ = \eta + \sqrt{\eta^2 + 2\frac{1-\alpha}{\kappa}}.$$

In the benchmark specification used in the project, we set $\alpha = 1/2$, so that

$$p_{AB}^- = -\eta + \sqrt{\eta^2 + \frac{1}{\kappa}}, \quad p_{AB}^+ = \eta + \sqrt{\eta^2 + \frac{1}{\kappa}}.$$

Therefore, the tail coefficients of f_t are

$$p_{t,AB}^- = \frac{p_{AB}^-}{\sigma_t \sqrt{t}}, \quad p_{t,AB}^+ = \frac{p_{AB}^+}{\sigma_t \sqrt{t}}.$$

For the Minimal Additive model, the normalized density is asymmetric Laplace. Therefore,

$$p_{MA}^- = \alpha_{MA}, \quad p_{MA}^+ = \beta_{MA},$$

and

$$p_{t,MA}^- = \frac{\alpha_{MA}}{\sigma_t \sqrt{t}}, \quad p_{t,MA}^+ = \frac{\beta_{MA}}{\sigma_t \sqrt{t}}.$$

For the Generalized Logistic model, the left and right tail coefficients are directly given by the two shape parameters:

$$p_{GL}^- = \alpha_{GL}, \quad p_{GL}^+ = \beta_{GL}.$$

Thus,

$$p_{t,GL}^- = \frac{\alpha_{GL}}{\sigma_t \sqrt{t}}, \quad p_{t,GL}^+ = \frac{\beta_{GL}}{\sigma_t \sqrt{t}}.$$

A useful consequence is that the time dependence of the tail coefficients enters only through the scale $\sigma_t \sqrt{t}$. For fixed shape parameters, a larger scale reduces both p_t^- and p_t^+ , making the exponential tails of f_t slower to decay.

0.2 Log-Linear Comparison of the PDFs

To visualize the tail behaviour described above, we compare the normalized PDFs of the three models on a log-linear scale. The comparison is performed on the normalized variable ζ , rather than directly on f_t . Since

$$f_t = \sigma_t \sqrt{t} \zeta_t.$$

moving from ζ_t to f_t only rescales the horizontal and vertical axes of the density. In particular, if $q_t = \sigma_t \sqrt{t}$, then

$$\rho_{f_t}(x) = \frac{1}{q_t} \rho_{\zeta_t} \left(\frac{x}{q_t} \right),$$

and the tail coefficients of f_t are obtained from the normalized ones by dividing by q_t .

For the graphical comparison, we choose common normalized tail coefficients

$$p^- = 0.9, \quad p^+ = 1.5.$$

For the MA and GL models this is obtained by setting

$$\alpha = p^-, \quad \beta = p^+.$$

For the Additive Bachelier benchmark, where $\alpha = 1/2$, we choose

$$\eta = \frac{p^+ - p^-}{2}, \quad \kappa = \frac{1}{p^+ p^-}.$$

Indeed, with these values,

$$p_{AB}^- = -\eta + \sqrt{\eta^2 + \frac{1}{\kappa}} = p^-, \quad p_{AB}^+ = \eta + \sqrt{\eta^2 + \frac{1}{\kappa}} = p^+.$$

Hence, the three models are compared under the same asymptotic tail decay. This makes the plot useful to highlight the different shapes of the densities around the center and in the transition from the body to the tails.

For MA and GL the normalized PDFs are available in closed form and are evaluated directly. For the Additive Bachelier model, instead, we reconstruct the PDF numerically from its characteristic function through Fourier inversion:

$$\rho_{AB}(y) = \frac{1}{2\pi} \int_{\mathbb{R}} e^{-iuy} \phi_{AB}(u) du.$$

In the implementation, this integral is computed by FFT on a uniform Fourier grid and the resulting density is interpolated on the target points.

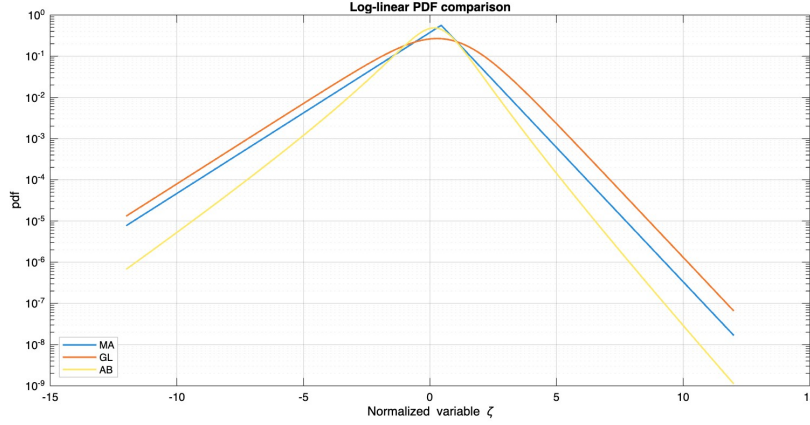


Figure 1: Log-linear comparison of the normalized PDFs for the MA, GL and AB models. The three models are plotted with common normalized tail coefficients $p^- = 0.9$ and $p^+ = 1.5$. The approximately linear behaviour in the tails confirms the exponential decay of the densities.

Figure 1 shows that all three models display exponential tail decay, as expected. Since the asymptotic coefficients are fixed to be the same, the main differences are visible in the central part of the distribution ($\zeta = 0$) and in the way each model connects the body of the density to its tails. The Minimal Additive (MA) model displays its characteristic cusp (lack of differentiability), which is an intrinsic property of the asymmetric Laplace distribution. The MA density transitions immediately from the central peak to the pure linear regime of the tails without a smooth intermediate zone. The Generalized Logistic (GL) model presents a distinctly smooth and rounded profile around the center and it allocates a significantly higher probability mass to the intermediate tails compared to the other two configurations, capturing a different distribution of medium-sized shocks. The Additive Bachelier (AB) while maintaining smoothness and differentiability around zero shows a more pronounced intermediate decay before converging to the common asymptotic tail regime.

1 Calibration of Forwards and Discount Factors

1.1 Synthetic Forwards

The first step of the project consists in extracting from market option prices the term structure of discount factors and forward prices. We follow the synthetic-forward methodology described in [2], which is based on the put-call parity relation for European options.

For a fixed maturity T and strike K , put-call parity gives

$$C(K, T) - P(K, T) = B(0, T) (F(0, T) - K),$$

where $C(K, T)$ and $P(K, T)$ are respectively the call and put prices, $B(0, T)$ is the discount factor and $F(0, T)$ is the forward price.

Defining the synthetic forward value as

$$G(K, T) = C(K, T) - P(K, T),$$

we obtain the linear relation

$$G(K, T) = -B(0, T)K + B(0, T)F(0, T).$$

Therefore, for each maturity, the market-implied discount factor and forward price can be recovered through a linear regression of $G(K, T)$ against the strike K .

Following the methodology described in [2], before running the regression we apply a liquidity filter by keeping only options with mid price larger than 0.1 index points. Then, call and put prices are matched only on common strikes. If fewer than three common liquid strikes are available, the corresponding maturity is discarded.

1.2 Market-Implied Discount Factors

For each valid maturity, the slope of the regression is equal to $-B(0, T)$, while the intercept is equal to $B(0, T)F(0, T)$. Hence, after estimating the regression coefficients, we recover

$$B(0, T) = -\text{slope},$$

and

$$F(0, T) = \frac{\text{intercept}}{B(0, T)}.$$

The first value in the original output is equal to NaN because the corresponding contract, JUN20, had already expired before the valuation date. In fact, JUN20 expires on the 14th May 2020, while the value date is the 2nd June 2020. Therefore, it is not included among the valid calibrated maturities. The calibrated discount factors are

$$B(0, T) = \begin{bmatrix} 0.9999 & 0.9998 & 0.9997 & 0.9997 & 0.9996 & 0.9999 & 1.0000 & 1.0000 \end{bmatrix}.$$

The quality of each regression is checked through the coefficient of determination R^2 as in [2]. The obtained values are

$$R^2 = \begin{bmatrix} 0.9999 & 0.9999 & 0.9999 & 0.9999 & 0.9999 & 0.9999 & 1.0000 & 1.0000 \end{bmatrix}.$$

The first six valid maturities have R^2 values extremely close to one, but not exactly equal to one. This means that the synthetic-forward relation is almost perfectly linear in the strike, although small residual deviations are still present. These deviations are expected in real market data, because option mid prices are affected by bid-ask spreads, liquidity effects, rounding errors and possible small inconsistencies across strikes.

The last two valid maturities instead have R^2 exactly equal to one. This indicates that, after the filtering and strike-matching procedure, the observed synthetic-forward values are perfectly aligned with a linear relation in the strike for those maturities. However, this result should not be over-interpreted. For longer maturities, the number of liquid common strikes is lower than for the maturities closer to the valuation date. As a consequence, a perfect R^2 may also be due to the small number of points used in the regression, rather than to a genuinely more robust calibration.

This point is relevant when interpreting the shape of the calibrated discount curve. In particular, the curve is not monotone: after decreasing for the first maturities, the calibrated discount factors increase again at the longest expiries. In a standard deterministic-rate framework, one would usually expect discount factors to be decreasing with maturity when rates are positive. Therefore, this upward movement should not be interpreted as a structural market feature without caution. Since the discount factors are obtained maturity by maturity from synthetic-forward regressions, this behaviour may reflect lower liquidity and the smaller number of common call-put strikes available at longer expiries. Hence, the far end of the calibrated curve is more uncertain than the short and medium maturities, even when the corresponding R^2 is equal to one.

Overall, the regression results confirm that the calibrated discount factors are broadly consistent with the option market quotes. Nevertheless, the reliability of the extracted market-implied discount curve is stronger for maturities close to the valuation date, where the calibration is supported by a richer set of common strikes, and weaker for the farthest maturities, where the regression is based on fewer observations.

Using the calibrated discount factors, we also compute continuously compounded zero rates as

$$r(0, T) = -\frac{\log B(0, T)}{T},$$

where the time to maturity T is computed using the ACT/365 convention as reported in [2]. Since all calibrated discount factors are very close to one, the resulting continuously compounded zero rates are close to zero. Moreover, the small increase of the discount factors at the longest maturities translates into very small, and possibly slightly negative, zero rates on the far end of the curve. This is consistent with the fact that the curve is directly implied from option prices and is not obtained through a smoothing or monotonicity-constrained bootstrap.

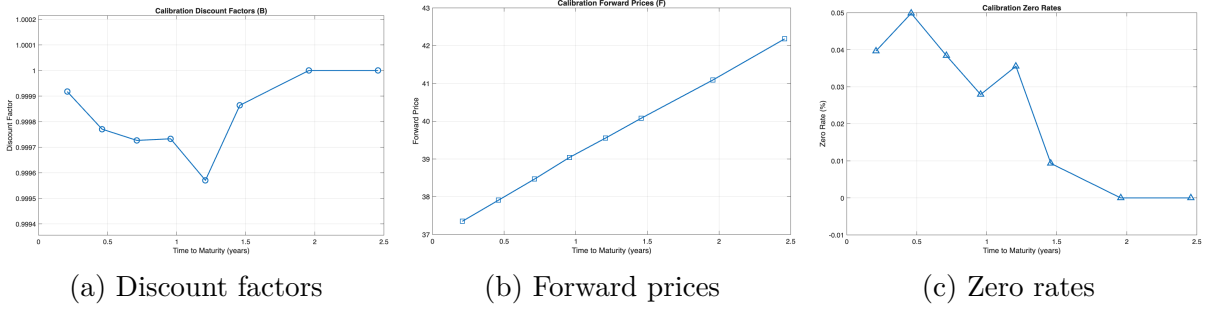


Figure 2: Calibrated discount factors, forward prices and continuously compounded zero rates.

1.3 Forward Curve and Absolute Dividends

The corresponding calibrated forward prices are

$$F(0, T) = \begin{bmatrix} 37.3505 & 37.9091 & 38.4689 & 39.0421 & 39.5549 & 40.0802 & 41.0900 & 42.1800 \end{bmatrix}.$$

These forwards are the market-implied forward prices obtained from option prices as shown in Section 1.2. They will be used in the next steps of the project as deterministic inputs common to the MA, GL and AB models.

Starting from the calibrated discount factors and forward prices, we infer the deterministic absolute dividends between consecutive maturities. For two consecutive expiries T_{j-1} and T_j , the absolute dividend capitalized at T_j is computed as

$$D_{j-1,j}^{T_j} = \frac{B(0, T_{j-1})}{B(0, T_j)} F(0, T_{j-1}) - F(0, T_j).$$

The corresponding piecewise-constant dividend intensity is

$$q_{j-1,j}^{abs} = \frac{D_{j-1,j}^{T_j}}{T_j - T_{j-1}}.$$

The calibrated absolute dividends between consecutive maturities are

$$D_{j-1,j}^{T_j} = \begin{bmatrix} -0.5531 & -0.5581 & -0.5735 & -0.5064 & -0.5370 & -1.0152 & -1.0900 \end{bmatrix}.$$

The corresponding absolute dividend intensities are

$$q_{j-1,j}^{abs} = \begin{bmatrix} -2.1943 & -2.2141 & -2.3521 & -2.0090 & -2.1539 & -2.0360 & -2.1740 \end{bmatrix}.$$

The negative sign means that the calibrated forward curve is increasing with maturity more than what is explained by discount-factor capitalization alone. In the linear additive setting of the project, this effect is interpreted as a deterministic absolute-dividend or

carry component. Therefore, from now on, the zero-rate curve and the absolute-dividend curve are treated as deterministic functions of time and kept fixed in the subsequent calibration, simulation and pricing steps.

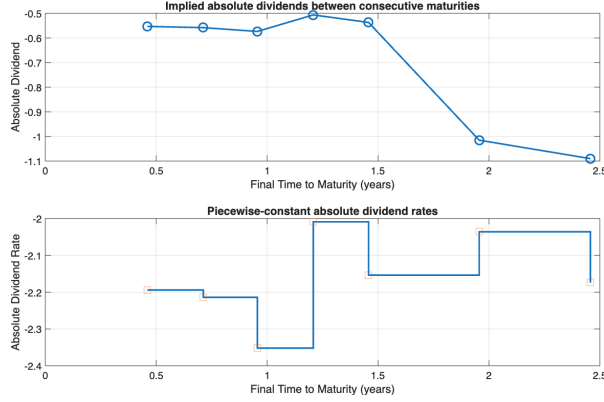


Figure 3: Implied absolute dividends and absolute dividend rates between consecutive maturities.

Overall, this first calibration step provides the deterministic market inputs required by the rest of the project: the discount curve $B(0, T)$, the forward curve $F(0, T)$, the zero-rate curve and the absolute-dividend curve. These quantities are model-independent.

2 Calibration of the Implied Volatility Surface

2.1 Models Description

The three models share the same structure $f_t = \sigma_t \sqrt{t} \zeta_t$ but differ in the distribution assumed for ζ_t , which determines the characteristic function $\phi_t(u) = \mathbb{E}_0[e^{iuf_t}]$. For each model, the characteristic function $\phi_t(u)$ is analytic in a horizontal strip of the complex plane of the form

$$\mathbb{S}_t = \{z \in \mathbb{C} : -p_t^+ < \text{Im}(z) < p_t^-\}, \quad (1)$$

Additive Bachelier (AB). The log-characteristic function is

$$\ln \phi_t^{AB}(u) = \psi \left(iu\eta\sigma_t\sqrt{t} + \frac{u^2}{2}\sigma_t^2 t; k, \frac{1}{2} \right) + iu\eta\sigma_t\sqrt{t}, \quad (2)$$

where

$$\psi(u; k, \alpha) = \frac{1}{k} \left[1 - \sqrt{1 + 2uk} \right] \quad (3)$$

is the Laplace exponent of the subordinator with $\alpha = 1/2$, $\eta \in \mathbb{R}$ is the skew parameter, and $k > 0$ is the vol-of-vol parameter.

Minimal Additive (MA). The distribution of ζ_t is an asymmetric Laplace with pdf

$$P_t(\zeta) = C_t \left[e^{\alpha_t(\zeta - \gamma_t)} \mathbf{1}_{\zeta < \gamma_t} + e^{-\beta_t(\zeta - \gamma_t)} \mathbf{1}_{\zeta \geq \gamma_t} \right], \quad (4)$$

where $C_t = \left(\frac{1}{\alpha_t} + \frac{1}{\beta_t} \right)^{-1}$ and $\gamma_t = \frac{1}{\alpha_t} - \frac{1}{\beta_t}$ is fixed by the martingality condition $\mathbb{E}_0[\zeta_t] = 0$. The characteristic function of f_t is:

$$\phi_t^{f,MA}(u) = e^{i\mu_t u} \frac{p_t^+ p_t^-}{(p_t^- + iu)(p_t^+ - iu)}, \quad (5)$$

where

$$\mu_t = \sigma_t \sqrt{t} \gamma_t. \quad (6)$$

Generalized Logistic (GL). The distribution of ζ_t is a Type IV Generalized Logistic (GL-IV) with pdf

$$P_t(\zeta) = \frac{\Gamma(\alpha_t + \beta_t)}{\Gamma(\alpha_t) \Gamma(\beta_t)} \frac{e^{\alpha_t(\zeta - \gamma_t)}}{(1 + e^{\zeta - \gamma_t})^{\alpha_t + \beta_t}}, \quad (7)$$

where γ_t is fixed by martingality. The log-characteristic function of f_t is:

$$\ln \phi_t^{GL}(u) = \ln \frac{\Gamma(\alpha_t + iu\sigma_t\sqrt{t}) \Gamma(\beta_t - iu\sigma_t\sqrt{t})}{\Gamma(\alpha_t) \Gamma(\beta_t)} + iu\sigma_t\sqrt{t} \gamma_t, \quad (8)$$

2.2 Additive Bachelier Calibration

Following the cascade calibration scheme proposed by [3], we calibrate the additive Bachelier model in three sequential stages, each targeting a different set of market instruments ordered by liquidity:

1. calibration of discount factors and ATM forwards from option prices;
2. estimation of the ATM volatility term structure σ_t^{ATM} ;
3. calibration of the skew parameter η and the vol-of-vol parameter k on the full OTM volatility surface.

The first stage was carried out in the previous section.

2.2.1 Stage 2: ATM Volatility Term Structure

In the second stage we estimate the ATM volatility σ_t^{ATM} for each available maturity. This quantity plays a central role in the model: by the separability property of the implied volatility (eq. (15) of [3]), the ATM vol is proportional to the model parameter σ_t via

$$\sigma_t^{ATM} = I_0 \sigma_t, \quad (9)$$

where I_0 depends only on η and k . Crucially, σ_t^{ATM} can be extracted directly from market data without any knowledge of η and k , which will only be determined in the third stage.

For each maturity, we compute the Bachelier implied volatility of each OTM option. Since the Bachelier vega is strictly positive, the implied volatility exists and is unique for any price strictly above the intrinsic value (Proposition 2.4 of [3]).

To estimate σ_t^{ATM} , we select the seven options with moneyness closest to zero and fit a quadratic polynomial to the pairs (x_i, σ_i^{impl}) . Evaluating the polynomial at $x = 0$ yields an estimate of the ATM implied volatility that accounts for the curvature of the smile and is more robust to noise in individual quotes than using a single option price. If options on both sides of the ATM are available, the quadratic fit is used; otherwise, the implied volatility of the nearest option is taken directly.

Finally, the normalized moneyness χ is computed for each option as

$$\chi = \frac{x}{\sigma_t^{ATM} \sqrt{t}}, \quad (10)$$

which will serve as the input for the third calibration stage.

2.2.2 Stage 3: Calibration of η and k

The third and final stage of the cascade calibration consists in estimating the two remaining parameters of the additive Bachelier model: the skew parameter η and the vol-of-vol parameter k . These parameters control, respectively, the asymmetry and the convexity of the implied volatility smile, and are calibrated jointly on the full OTM volatility surface.

We first build the OTM surface by selecting, for each maturity, only the out-of-the-money options: calls with positive moneyness ($x > 0$) and puts with negative moneyness ($x < 0$). Put prices are converted into call-equivalent prices using put-call parity in the Bachelier framework:

$$C = P - B_0 x, \quad (11)$$

so that all o

Denoting by $C(x, t, \sigma_t, \eta, k)$ the additive Bachelier model call price as a function of moneyness x , time to maturity t , and model parameters (σ_t, η, k) , once we impose $\sigma_t = \frac{\sigma_t^{ATM}}{I_0}$, the model price is a function of only η and k :

$$C^{mod}(x, t, \eta, k) := C\left(x, t, \sigma_t = \frac{\sigma_t^{ATM}}{I_0}, \eta, k\right) = B_0 \sigma_t^{ATM} \sqrt{t} \Psi(\chi, \eta, k) \quad (12)$$

where

$$\Psi(\chi, \eta, k) := \frac{1}{I_0} \mathbb{E}\left[c^b(\chi I_0 + \eta(G - 1), \sqrt{G})\right] = \frac{1}{I_0 B_0} C(\chi I_0, 1, 1, \eta, k) \quad (13)$$

$$I_0 = \sqrt{2\pi} \mathbb{E}_0 \left[c^b \left(\eta(G-1), \sqrt{G} \right) \right] = \frac{\sqrt{2\pi}}{B_0} C(0, 1, 1, \eta, k) \quad (14)$$

The last stage requires minimizing the L^2 distance between the quoted market prices $C^{mkt}(x_i, t_i)$ and the corresponding model prices:

$$\begin{aligned} d(\eta, k) &:= \sum_i \left[C^{mkt}(x_i, t_i) - C^{mod}(x_i, t_i; \eta, k) \right]^2 \\ &= \sum_i \left[C^{mkt}(\chi_i \sigma_{t_i}^{ATM} \sqrt{t_i}, t_i) - B_0 \sigma_{t_i}^{ATM} \sqrt{t_i} \Psi(\chi_i, \eta, k) \right]^2 \end{aligned} \quad (15)$$

We evaluate $C(x, t, \sigma, \eta, k)$ with equation (8) from [3] via the FFT algorithm and choosing the Lewis shift a inside the analyticity strip as $a = -0.5 p^+$.

The minimization is performed via `fminsearch` in MATLAB, starting from an initial guess $(\eta_0, k_0) = (0.2, 1)$. The optimizer returns the optimal parameters $\hat{\eta}$ and $\hat{k}$.

Thanks to the separability property of the implied volatility (eq. (18)), the function $\Psi(\chi; \eta, k)$ does not depend on the maturity t nor on σ_t^{ATM} . This means that a single pair $(\hat{\eta}, \hat{k})$ is calibrated globally across all maturities, making the optimization particularly parsimonious: only one FFT evaluation per candidate (η, k) is required to price all options on the surface simultaneously.

Table 1: Calibrated parameters of the Additive Bachelier model.

$\hat{\eta}$	$\hat{k}$
-0.0624	1.0180

The skew parameter $\hat{\eta} = -0.0624$ is small and negative, indicating a mild left skew in the distribution of f_t : the left tail is slightly heavier than the right tail. The vol-of-vol parameter $\hat{k} = 1.0180$ is close to one, indicating a moderate degree of convexity in the implied volatility smile: larger values of k would produce a more pronounced smile, while $k \rightarrow 0$ would collapse the model to the standard Bachelier framework with a flat smile.

2.3 Minimal Additive and Generalized Logistic Calibration

We now address whether the cascade calibration scheme described in [3], Sec. 3 can be applied also to the Minimal Additive (MA) and Generalized Logistic (GL) models when the tail parameters $\alpha_t = \alpha$ and $\beta_t = \beta$ do not depend on time. The answer is affirmative. The key observation is that, in both models, the forward price is modeled as

$$F(t, T) = F(0, T) + f_t, \quad f_t = \sigma_t \sqrt{t} \zeta_t, \quad (16)$$

where the shape of the distribution of ζ_t is controlled by α and β (η and κ in the AB model) only, while σ_t controls the scale. Thanks to this, defining $x = K - F_0$, we can write:

$$\begin{aligned} C(x, t, \sigma_t, \alpha, \beta) &= B_0 \mathbb{E}_0[f_t - x]^+ \\ &= B_0 \sigma_t \sqrt{t} \mathbb{E}_0 \left[\underbrace{\frac{f_t}{\sigma_t \sqrt{t}}}_{\zeta_t} - \underbrace{\frac{x}{\sigma_t \sqrt{t}}}_y \right]^+ \\ &= B_0 \sigma_t \sqrt{t} \mathbb{E}_0[\zeta_t - y]^+ = B_0 \sigma_t \sqrt{t} c^{\text{norm}}(y; \alpha, \beta), \end{aligned} \quad (17)$$

where $c^{\text{norm}}(y; \alpha, \beta) = C(y, 1, 1, \alpha, \beta)/B_0$ is the normalized call price, which depends only on α and β : separability of the implied volatility still holds and it enables the same three-stage cascade calibration used for the additive Bachelier model.

The ATM call price evaluated at $x = 0$ gives:

$$C^{\text{mkt}}(x = 0, t) = B_0 \sigma_t \sqrt{t} \cdot c^{\text{norm}}(0; \alpha, \beta). \quad (18)$$

Defining

$$I_0(\alpha, \beta) := \sqrt{2\pi} c^{\text{norm}}(0; \alpha, \beta), \quad (19)$$

the ATM volatility satisfies $\sigma_t^{\text{ATM}} = I_0 \sigma_t$, exactly as in the additive Bachelier model. Therefore, once σ_t^{ATM} has been extracted from market data in Stage 2, which is model-independent and uses the same procedure for all three models, the normalized moneyness

$$\chi = \frac{x}{\sigma_t^{\text{ATM}} \sqrt{t}} \quad (20)$$

can be computed for each option. The model price then takes the form:

$$C^{\text{mod}}(x, t; \alpha, \beta) = B_0 \sigma_t^{\text{ATM}} \sqrt{t} \frac{c^{\text{norm}}(\chi I_0; \alpha, \beta)}{I_0}, \quad (21)$$

which depends on t and σ_t^{ATM} only through the overall scale $B_0 \sigma_t^{\text{ATM}} \sqrt{t}$, while the shape function $c^{\text{norm}}(\chi I_0; \alpha, \beta)/I_0$ depends only on α and β .

2.3.1 Implementation for the MA Model

For the MA model, the normalized call price admits a closed-form expression:

$$c^{\text{norm}}(y; \alpha, \beta) = \begin{cases} \frac{C}{\alpha^2} e^{\alpha(y-\gamma)} - y & y < \gamma \\ \frac{C}{\beta^2} e^{-\beta(y-\gamma)} & y \geq \gamma \end{cases} \quad (22)$$

where $C = \left(\frac{1}{\alpha} + \frac{1}{\beta}\right)^{-1}$ and $\gamma = \frac{1}{\alpha} - \frac{1}{\beta}$ is the localization parameter imposed by the martingality condition $\mathbb{E}_0[\zeta_t] = 0$. The parameters α and β are calibrated by minimizing the L^2 distance between market and model prices:

$$d(\alpha, \beta) := \sum_i \left[C_i^{mkt} - B_0 \sigma_{t_i}^{ATM} \sqrt{t_i} \frac{c^{\text{norm}}(\chi_i I_0; \alpha, \beta)}{I_0} \right]^2. \quad (23)$$

Identifiability Issue in the MA Model

When calibrating the MA model, we observe that the optimal values of α and β depend significantly on the initial guess provided to the optimizer, while their ratio α/β remains stable across different starting points. This behaviour signals a numerical identifiability issue that we now explain.

In principle, α and β are separately identifiable since they control the exponential decay rate of the left and right tails respectively. However, in practice the quoted OTM options cover only a limited range of moneyness, so that the objective function is very insensitive to the absolute values of α and β individually: many pairs (α, β) with the same ratio but different absolute values produce nearly identical call prices over the observed moneyness range, creating a numerically flat valley in the parameter space.

The ratio α/β is instead well identified because it controls the *asymmetry* between the left and right tails, which is directly observable from the skewness of the implied volatility smile.

To handle this issue, we run the optimizer from a grid of initial points and retain the solution closest to unit scale, namely $\alpha^* = 1.0239$ and $\beta^* = 0.9980$, to avoid introducing an arbitrary extreme scaling.

α_0	β_0	α^*	β^*	α^*/β^*	Objective
0.50	0.50	0.5120	0.4990	1.0260	10.6299
0.50	1.00	0.6526	0.6360	1.0260	10.6299
0.50	2.00	0.7968	0.7766	1.0260	10.6299
0.50	5.00	0.8519	0.8303	1.0260	10.6299
1.00	0.50	0.6644	0.6476	1.0260	10.6299
1.00	1.00	1.0239	0.9980	1.0260	10.6299
1.00	2.00	1.3051	1.2720	1.0260	10.6299
1.00	5.00	1.5969	1.5565	1.0260	10.6299
2.00	0.50	0.7983	0.7781	1.0260	10.6299
2.00	1.00	1.3289	1.2952	1.0260	10.6299
2.00	2.00	2.0479	1.9960	1.0260	10.6299
2.00	5.00	2.8301	2.7584	1.0260	10.6299
5.00	0.50	0.8762	0.8540	1.0260	10.6299
5.00	1.00	1.6367	1.5952	1.0260	10.6299
5.00	2.00	2.8769	2.8040	1.0260	10.6299
5.00	5.00	5.1218	4.9920	1.0260	10.6299

Table 2: MA model calibration results for different initial conditions (α_0, β_0) . The calibrated values α^* and β^* vary substantially across starting points, while their ratio α^*/β^* remains constant at approximately 1.0260. The objective function is also unchanged across all runs, indicating that the calibration identifies the relative tail asymmetry but not the common scale of the two tail coefficients.

2.3.2 Implementation for the GL Model

The calibration of the Generalized Logistic (GL) model follows exactly the same cascade calibration procedure described for the Additive Bachelier model. The only difference lies in the characteristic function used to evaluate the normalized call price via the Bachelier-Lewis FFT formula, which requires the evaluation of the log-Gamma function $\ln \Gamma(z)$ at complex arguments $z \in \mathbb{C}$.

Since MATLAB's built-in `gamma` function is only defined for real arguments, we implemented a dedicated routine `generalizedLogisticLogCf.m` that evaluates $\ln \Gamma(z)$ in the complex plane via the Lanczos approximation in logarithmic form.

Table 3: Calibrated parameters of the Generalized Logistic model.

$\hat{\alpha}$	$\hat{\beta}$
1.0180	0.3679

2.4 Model Comparison

Table 4: Global calibration quality: RMSE of model call-equivalent prices across the full OTM surface (596 options).

	AB	MA	GL
RMSE	0.1007	0.1336	0.1052

Table 5: Calibration RMSE by maturity.

Maturity	N	RMSE_{AB}	RMSE_{MA}	RMSE_{GL}
17-Aug-2020	188	0.0941	0.0822	0.1005
17-Nov-2020	204	0.0637	0.0927	0.0714
17-Feb-2021	38	0.0989	0.1767	0.0965
17-May-2021	44	0.0974	0.1859	0.0964
17-Aug-2021	13	0.0892	0.1914	0.0866
16-Nov-2021	69	0.0674	0.1461	0.0704
17-May-2022	11	0.1120	0.0913	0.1291
16-Nov-2022	29	0.2759	0.3141	0.2798

Table 6: Calibration RMSE by normalised moneyness bucket χ .

Moneyness bucket	N	RMSE_{AB}	RMSE_{MA}	RMSE_{GL}
$\chi < -2$	56	0.0780	0.0627	0.0885
$-2 \leq \chi < -1$	49	0.1183	0.1730	0.1196
$-1 \leq \chi < 0$	82	0.1441	0.2167	0.1427
$0 \leq \chi < 1$	116	0.1191	0.1556	0.1199
$1 \leq \chi < 2$	86	0.1021	0.1342	0.1067
$\chi \geq 2$	207	0.0611	0.0596	0.0741

The calibration quality is assessed through the RMSE of model call-equivalent prices across the full OTM surface, broken down globally, by maturity, and by normalised moneyness bucket.

At the global level (Table 4), the AB and GL models achieve comparable fit quality, with RMSE of 0.1007 and 0.1052 respectively, while the MA model shows a slightly higher RMSE of 0.1336. This is consistent with the greater flexibility of the AB and GL characteristic functions in reproducing the shape of the implied volatility surface.

The breakdown by maturity (Table 5) reveals that the largest errors are concentrated at the longest expiry (16-Nov-2022), where all three models exhibit significantly higher RMSE. This is consistent with the lower liquidity observed at long maturities, which makes the calibration less reliable due to the smaller number of common liquid strikes available.

The breakdown by moneyness bucket (Table 6) shows that the fit deteriorates for near-ATM options ($-1 \leq \chi < 1$), where the MA model in particular displays the highest RMSE across all buckets. This suggests that the asymmetric Laplace density, which transitions immediately from its central cusp to the exponential tail regime without a smooth intermediate zone, struggles to reproduce the body of the implied volatility surface as accurately as the AB and GL models. Deep OTM options ($|\chi| \geq 2$) are fitted well by all three models, reflecting the fact that exponential tail behaviour is a structural feature shared by all three distributional assumptions.

3 Monte Carlo Simulation

3.1 Characteristic Function of the Increments

A key property of additive processes is that they have independent increments. This allows us to compute the characteristic function of the increment $f_t - f_s$ for any pair of reset dates $t > s \geq t_0$ using the additivity property:

$$\mathbb{E}_0[e^{iuf_t}] = \mathbb{E}_0[e^{iu(f_t - f_s)}] \mathbb{E}_0[e^{iuf_s}], \quad (24)$$

which immediately gives:

$$\phi_{s,t}(u) = \frac{\phi_t(u)}{\phi_s(u)}. \quad (25)$$

Additive Bachelier (AB)

The characteristic function of the increment for the AB model is:

$$\phi_{s,t}^{AB}(u) = \frac{\phi_t^{AB}(u)}{\phi_s^{AB}(u)} = \exp(\ln \phi_t^{AB}(u) - \ln \phi_s^{AB}(u)), \quad (26)$$

which gives explicitly:

$$\begin{aligned} \ln \phi_{s,t}^{AB}(u) = & \frac{1}{k} \left[\sqrt{1 + 2k \left(iu\eta\sigma_s\sqrt{s} + \frac{u^2}{2}\sigma_s^2s \right)} - \sqrt{1 + 2k \left(iu\eta\sigma_t\sqrt{t} + \frac{u^2}{2}\sigma_t^2t \right)} \right] \\ & + iu\eta(\sigma_t\sqrt{t} - \sigma_s\sqrt{s}). \end{aligned} \quad (27)$$

Minimal Additive (MA)

For the MA model, using the closed-form expression for $\phi_t^{f,MA}$, the characteristic function of the increment is:

$$\phi_{s,t}^{MA}(u) = \frac{\phi_t^{f,MA}(u)}{\phi_s^{f,MA}(u)} = e^{i(\mu_t - \mu_s)u} \frac{p_t^+ p_t^-}{p_s^+ p_s^-} \cdot \frac{(p_s^- + iu)(p_s^+ - iu)}{(p_t^- + iu)(p_t^+ - iu)}. \quad (28)$$

Defining

$$q_0 = \frac{p_t^+ p_t^-}{p_s^+ p_s^-}, \quad (29)$$

$$A^- = q_0 \frac{(p_s^- - p_t^-)(p_s^+ + p_t^-)}{p_t^+ + p_t^-} \quad (\text{left tail}), \quad (30)$$

$$A^+ = q_0 \frac{(p_s^- + p_t^+)(p_s^+ - p_t^+)}{p_t^+ + p_t^-} \quad (\text{right tail}), \quad (31)$$

$$d = \mu_t - \mu_s, \quad (32)$$

we can rewrite the characteristic function in partial fraction form as:

$$\phi_{s,t}^{MA}(u) = e^{iud} \left\{ q_0 + \frac{A^-}{p_t^- + iu} + \frac{A^+}{p_t^+ - iu} \right\}. \quad (33)$$

The partial fraction representation has a useful probabilistic interpretation. The two rational terms vanish as $|u| \rightarrow \infty$, while the term $q_0 e^{iud}$ does not vanish: it only oscillates with constant modulus q_0 . This is the signature of a point mass in the increment distribution. Indeed, the characteristic function of a deterministic random variable equal to d is e^{iud} . Therefore, the MA increment has an atom located at $d = \mu_t - \mu_s$ with probability

$$\mathbb{P}(f_t - f_s = d) = q_0 = \frac{p_t^- p_t^+}{p_s^- p_s^+}. \quad (34)$$

The remaining rational terms correspond to the absolutely continuous component of the distribution.

Generalized Logistic (GL)

The characteristic function of the increment for the GL model is:

$$\phi_{s,t}^{GL}(u) = \frac{\phi_t^{GL}(u)}{\phi_s^{GL}(u)} = \exp(\ln \phi_t^{GL}(u) - \ln \phi_s^{GL}(u)). \quad (35)$$

In our implementation, with time-independent parameters $\alpha_t = \alpha$ and $\beta_t = \beta$, this simplifies to:

$$\begin{aligned} \ln \phi_{s,t}^{GL}(u) = & \ln \frac{\Gamma(\alpha + iu\sigma_t\sqrt{t})}{\Gamma(\alpha + iu\sigma_s\sqrt{s})} + \ln \frac{\Gamma(\beta - iu\sigma_t\sqrt{t})}{\Gamma(\beta - iu\sigma_s\sqrt{s})} \\ & + iu\gamma(\sigma_t\sqrt{t} - \sigma_s\sqrt{s}). \end{aligned} \quad (36)$$

Analyticity Strip of the Increment CF

A fundamental structural property shared by all three models is that the analyticity strip of the increment characteristic function $\phi_{s,t}(u)$ is determined solely by the parameters at the final maturity t . Formally,

$$\mathbb{S}_{s,t} = \left\{ z \in \mathbb{C} : -p_t^+ < \text{Im}(z) < p_t^- \right\}, \quad (37)$$

where $p_t^\pm$ are the same boundaries defined for the marginal CF ϕ_t in Section 2.1. This follows from the ratio $\phi_{s,t} = \phi_t/\phi_s$: since the admissibility of additive processes requires $p_\tau^\pm$ to be non-increasing in τ , we have $p_t^\pm \leq p_s^\pm$, so the singularities of ϕ_t always lie strictly closer to the real axis than those of ϕ_s . The denominator ϕ_s therefore introduces no additional singularities within $\mathbb{S}_{s,t}$, and the final maturity t acts as the sole bottleneck for analyticity. This uniform structure justifies applying the same complex shift a in the FFT inversion across all three models.

3.2 AB and GL Model Increment Simulation

To simulate the increments in the AB and GL cases, we implemented Algorithm 1 of [5]. The procedure consists of two main stages: retrieving the CDF of the increment via FFT, and inverting it to generate samples.

3.2.1 Retrieving the CDF via FFT

Let $X_{s,t} := f_t - f_s$ denote the increment between two reset dates $s < t$, with CDF

$$P_{s,t}(x) := \mathbb{P}(X_{s,t} \leq x) = \mathbb{E}_0[\mathbf{1}_{\{X_{s,t} \leq x\}}].$$

The CDF is recovered by Fourier inversion of the characteristic function $\phi_{s,t}(u)$. Rather than applying the standard inversion formula directly, which presents a singularity at the origin, we follow the Lewis-FFT approach and shift the integration contour into the complex plane by a parameter $a \neq 0$ chosen inside the analyticity strip $\mathbb{S}_{s,t} = \{z \in \mathbb{C} :$

$-p_t^+ < \text{Im}(z) < p_t^-$ of $\phi_{s,t}$. The CDF is then computed as

$$P_{s,t}(x) = R_a - \frac{e^{-ax}}{2\pi} \int_{-\infty}^{+\infty} e^{-iux} \frac{\phi_{s,t}(u - ia)}{i(u - ia)} du, \quad (38)$$

where

$$R_a = \begin{cases} 1, & 0 < a < p_t^+, \\ 0, & -p_t^- < a < 0, \end{cases} \quad (39)$$

The complex shift removes the pole at the origin and stabilises the numerical inversion. In the implementation, the integral is discretised on an equally spaced Fourier grid of $N = 2^M$ points with step size du and computed via FFT, producing a numerical approximation $\hat{P}_{s,t}(x)$ on a finite x -grid covering $[x_1, -x_1]$ with spatial step $dx = -2x_1/(N - 1)$, linked to the Fourier step by the relation $du = 2\pi/(N \cdot dx)$.

3.2.2 CDF Restriction, Interpolation, and Tail Extrapolation

The raw FFT output may contain spurious oscillations or non-monotone regions near the boundaries of the grid. We therefore restrict the x -grid to the largest consecutive set of points $\{x_n\}$ for which $\hat{P}_{s,t}$ is strictly increasing and lies in $[0, 1]$. We denote the left and right endpoints of this restricted interval by x_b and x_e , respectively, so that the valid region is $[x_b, x_e]$.

Within $[x_b, x_e]$, the CDF is approximated by a cubic spline interpolant built on the grid values $\hat{P}(x_b), \dots, \hat{P}(x_e)$. This spline is then used to invert the CDF: given a uniform draw $U \sim \mathcal{U}(0, 1)$ with $\hat{P}(x_b) \leq U \leq \hat{P}(x_e)$, the sample is obtained as $X_{s,t} = \text{spline}^{-1}(U)$.

Outside the range $[x_b, x_e]$, the CDF must be extrapolated. In our first implementation, we followed the standard approach of Algorithm 1 of [5], estimating the tail decay rates empirically from the boundary values of the restricted grid:

$$\lambda_- := \frac{\ln \hat{P}(x_{b+1}) - \ln \hat{P}(x_b)}{x_{b+1} - x_b}, \quad \lambda_+ := \frac{\ln \hat{P}(x_e) - \ln \hat{P}(x_{e-1})}{x_e - x_{e-1}}. \quad (40)$$

However, this approach produced unsatisfactory results in practice, as confirmed by the numerical comparison reported in Table 7. The table reveals two distinct failure modes of the empirical estimator. For the right tail, the estimated rate $\hat{\lambda}_+$ collapses to essentially zero ($\approx 5 \times 10^{-9}$) for all three models, while the theoretical value p_t^+ is of order 10^{-2} – 10^{-1} . This occurs because near x_e the reconstructed CDF $\hat{P}$ is already extremely close to one. For the left tail, the picture is model-dependent: in the AB and GL cases the estimated $\hat{\lambda}_-$ is significantly smaller than the theoretical p_t^- , while in the MA case it is larger. In both situations the empirical estimate is unreliable, as the boundary region of the restricted grid does not contain sufficient information to identify the true exponential decay rate.

Table 7: Comparison between empirically estimated and theoretical tail decay rates for the increment CDF at the reset dates $T_1 = 6$ m, $T_2 = 1$ y.

Model	$\hat{\lambda}_-$ (est.)	p_t^- (theor.)	$\hat{\lambda}_+$ (est.)	p_t^+ (theor.)
AB	3.37×10^{-3}	6.60×10^{-2}	5.47×10^{-9}	5.82×10^{-2}
GL	2.23×10^{-2}	1.01×10^{-1}	5.46×10^{-9}	9.14×10^{-2}
MA	1.58×10^{-1}	8.69×10^{-2}	5.46×10^{-9}	8.47×10^{-2}

To address this, we adopted the theoretically grounded extrapolation suggested in [5], which uses the analyticity strip boundaries p_t^- and p_t^+ as the true asymptotic decay rates of the CDF tails. Since these values are available in closed form for all three models from the characteristic function analyticity strip $\mathbb{S}_{s,t}$, they carry no estimation uncertainty and are by construction consistent with the model specification. The extrapolation formula is:

$$\hat{P}(x) = \begin{cases} \hat{P}(x_b) e^{p_t^-(x-x_b)} & \text{for } x \leq x_b, \\ 1 - (1 - \hat{P}(x_e)) e^{-p_t^+(x-x_e)} & \text{for } x \geq x_e, \end{cases} \quad (41)$$

and the corresponding inverse CDF in the tails is:

$$X_{s,t} = \begin{cases} x_b + \frac{\ln(U/\hat{P}(x_b))}{p_t^-} & \text{for } U < \hat{P}(x_b), \\ x_e - \frac{\ln((1-U)/(1-\hat{P}(x_e)))}{p_t^+} & \text{for } U > \hat{P}(x_e). \end{cases} \quad (42)$$

Adopting this theoretically grounded extrapolation significantly improved the accuracy of the simulated paths, yielding well-behaved CDF inversions across all monitored dates and strike levels.

Finally, the conditional CDF of f_t given $f_s = y$ is obtained by a simple shift: $\mathbb{P}(f_t \leq z \mid f_s = y) = P_{s,t}(z - y)$. Thus, reconstructing the CDF of the increment is sufficient to simulate the process between any two reset dates.

3.3 MA Model: Simulation with Probability Atom

As shown above, the increment CF $\phi_{s,t}^{MA}(u)$ admits the partial fraction decomposition

$$\phi_{s,t}^{MA}(u) = e^{iud} \left\{ q_0 + \frac{A^-}{p_t^- + iu} + \frac{A^+}{p_t^+ - iu} \right\}, \quad (43)$$

where the term $q_0 e^{iud}$ signals the presence of a probability atom of mass q_0 located at $d = \mu_t - \mu_s$, while the two rational terms constitute the absolutely continuous component. Since $q_0 e^{iud}$ does not vanish as $|u| \rightarrow \infty$, the full CF $\phi_{s,t}^{MA}$ does not satisfy the decay

condition required for stable FFT-based CDF inversion. Passing the full CF directly to the Lewis-FFT-S algorithm would therefore produce an inaccurate result.

To handle this, we follow the approach of [5] for finite-activity processes, and decompose the increment distribution into its atom and its absolutely continuous part. The CF of the continuous component, conditional on the atom not being realised, is obtained by subtracting the atom's contribution and renormalising:

$$\phi_{s,t}^{\mathbb{C}}(u) = \frac{\phi_{s,t}^{MA}(u) - q_0 e^{iud}}{1 - q_0} = \frac{e^{iud}}{1 - q_0} \left\{ \frac{A^-}{p_t^- + iu} + \frac{A^+}{p_t^+ - iu} \right\}. \quad (44)$$

By construction, $\phi_{s,t}^{\mathbb{C}}(u) \rightarrow 0$ as $|u| \rightarrow \infty$, so the Lewis-FFT-S algorithm described in subsection 3.2 can be applied to it directly.

The simulation of each increment $X_{s,t} = f_t - f_s$ then proceeds in two steps:

1. Draw $B \sim \text{Bernoulli}(1 - q_0)$.
2. If $B = 0$: set $X_{s,t} = d$ (the atom is realised).
 If $B = 1$: simulate $X_{s,t}$ from the absolutely continuous component by applying Algorithm 1 of [5] to $\phi_{s,t}^{\mathbb{C}}(u)$

The special case $s = 0$ is handled separately. When $s = 0$, we have $q_s^\pm = p_s^\pm \rightarrow \infty$ and therefore $q_0 \rightarrow 0$: the increment f_t is absolutely continuous and no atom is present. The simulation reduces directly to the standard Lewis-FFT-S inversion of the full CF $\phi_{0,t}^{MA}(u)$, with no Bernoulli draw required.

3.4 FFT Parameter Selection and Numerical Error Control

3.4.1 Sources of Numerical Error

The FFT reconstruction of the CDF introduces two main numerical errors. The first one is the *truncation error*, due to replacing the Fourier integral over the whole real line by an integral over a finite frequency range. This error is controlled by the size of the Fourier domain and by the decay of the shifted characteristic function.

The second one is the *discretisation error*, due to approximating the Fourier integral by a finite sum on an equally spaced grid. This error depends on the Fourier step size and on the number of FFT points. The complex shift a , chosen inside the analyticity strip, is used not only to remove the singularity at the origin but also to improve the balance between truncation and discretisation errors, as discussed in [1] and [5].

In practice, these errors are controlled by using a sufficiently large FFT grid and by verifying that the probability mass left outside the valid FFT range is negligible.

3.4.2 Parameter Selection

The FFT grid is characterised by three parameters: M (grid size), x_1 (left boundary of the x -grid), and a (complex shift). We set $N = 2^M$ with $M = 15$, yielding $N = 32\,768$ grid points.

Grid range. The x -grid is chosen symmetric around zero, covering the interval $[x_1, -x_1]$ with $x_1 = -300$. This choice is motivated by the range of strikes observed in the dataset: the largest absolute out-of-the-money moneyness level is approximately 162, so the grid extends well beyond the relevant support of the increment distribution, providing a safety margin of approximately 138 units on each side. The corresponding spatial and Fourier steps are

$$dx = \frac{-2x_1}{N-1} \approx 1.83 \times 10^{-2}, \quad du = \frac{2\pi}{N \cdot dx} \approx 1.05 \times 10^{-2}. \quad (45)$$

Complex shift. Following the rule of thumb of [5], the complex shift is chosen as $a = p_t^+/2$, placing the integration path approximately midway between the origin and the right boundary of the analyticity strip $\mathbb{S}_{s,t}$. This choice balances the truncation and discretisation errors, minimising the total error bound for a given grid size N .

3.4.3 Tail condition

As a practical validation of the grid adequacy, we verify that the reconstructed CDF satisfies

$$\max(\hat{P}(x_b), 1 - \hat{P}(x_e)) \leq 10^{-4}, \quad (46)$$

where $[x_b, x_e]$ is the valid region of the restricted grid. This condition, suggested in [5], ensures that negligible probability mass lies outside the grid boundaries, so that the exponential tail extrapolation introduces no appreciable error.

Table 8: Probability mass outside the valid FFT domain for each model. TailMass = $\max(\hat{P}(x_b), 1 - \hat{P}(x_e))$.

Model	$\hat{P}(x_b)$	$1 - \hat{P}(x_e)$	TailMass
AB	1.62×10^{-6}	5.90×10^{-8}	1.62×10^{-6}
GL	2.45×10^{-7}	5.97×10^{-8}	2.45×10^{-7}
MA	6.51×10^{-6}	6.45×10^{-8}	6.51×10^{-6}

3.5 Forward Start Call Option

In this section we verify the adequacy of the Monte Carlo simulation on the forward start call option with $T_1 = 6m$ and $T_2 = 1y$, for a given strike K_2 , whose payoff at T_2 is

$$\left[S_{T_2} - K F(T_1, T_2) \right]^+. \quad (47)$$

Following the notation of the Forward document [6], the forward price at T_1 seen from T_2 and the spot price at T_2 are given by

$$F(T_1, T_2) = F(t_0, T_2) + f_{T_1, T_2}, \quad (48)$$

$$S_{T_2} = F(T_2, T_2) = F(t_0, T_2) + f_{T_2, T_2}, \quad (49)$$

where f_{T_1, T_2} and f_{T_2, T_2} denote the additive process evaluated at the relevant dates.

We use the relation (Lemma 2 of [6]):

$$f_{T_1, T_2} = e^{\int_{T_1}^{T_2} r_s ds} f_{T_1, T_1} = \frac{B(t_0, T_1)}{B(t_0, T_2)} f_{T_1, T_1}, \quad (50)$$

and the decomposition of the increment:

$$f_{T_2, T_2} = f_{T_1, T_1} + X_{T_1, T_2}, \quad (51)$$

where $X_{T_1, T_2} := f_{T_2, T_2} - f_{T_1, T_1}$ is the increment of the additive process between T_1 and T_2 .

For each simulated path, we proceed as follows:

1. Simulate $f_{T_1, T_1} = X_{T_1}$ from the marginal characteristic function $\phi_{T_1}(u)$, using the Lewis-FFT-S algorithm described in Section 3.
2. Simulate the increment X_{T_1, T_2} from the increment characteristic function $\phi_{T_1, T_2}(u)$, using the same algorithm (with the Bernoulli decomposition in the MA case).
3. Reconstruct the full path value at T_2 :

$$X_{T_2} = X_{T_1} + X_{T_1, T_2}. \quad (52)$$

For each simulated path $j = 1, \dots, N_{\text{sim}}$, the relevant quantities are:

$$S_{T_2}^{(j)} = F(t_0, T_2) + X_{T_2}^{(j)}, \quad (53)$$

$$K_2 F(T_1, T_2)^{(j)} = K_2 \left(F(t_0, T_2) + \frac{B(t_0, T_1)}{B(t_0, T_2)} X_{T_1}^{(j)} \right), \quad (54)$$

$$\text{Payoff}^{(j)} = \max \left(S_{T_2}^{(j)} - K_2 F(T_1, T_2)^{(j)}, 0 \right). \quad (55)$$

Finally, the price at t_0 is obtained by discounting the average payoff:

$$\text{Price}(t_0) = B(t_0, T_2) \frac{1}{N_{\text{sim}}} \sum_{j=1}^{N_{\text{sim}}} \text{Payoff}^{(j)}. \quad (56)$$

Table 9: Monte Carlo prices of the forward start call option with $T_1 = 6$ m, $T_2 = 1$ y and $K_2 = 1$, $N_{\text{sim}} = 5000000$, under the three calibrated models. The 95% confidence interval is $[\hat{\pi} - 1.96 \text{ SE}, \hat{\pi} + 1.96 \text{ SE}]$.

Model	MC Price	MC Std. Error	CI ₉₅ Low	CI ₉₅ High
AB	2.9819	2.98×10^{-3}	2.9760	2.9877
GL	2.9261	2.96×10^{-3}	2.9203	2.9319
MA	2.2942	3.12×10^{-3}	2.2881	2.3004

3.5.1 Analytical MA Forward-Start Benchmark

For the Minimal Additive model, the forward-start call can also be priced analytically. This provides a useful benchmark for the Monte Carlo simulation. Let

$$X_{T_1} := f_{T_1, T_1}, \quad X_{T_1, T_2} := f_{T_2, T_2} - f_{T_1, T_1}.$$

By additivity, X_{T_1, T_2} is independent of X_{T_1} . Using the relation between the forward values and the additive shocks, the payoff can be written as

$$\left[X_{T_1, T_2} + (1 - K_2)F(0, T_2) + \left(1 - \frac{K_2}{B(T_1, T_2)} \right) X_{T_1} \right]^+.$$

Equivalently, conditional on $X_{T_1} = x$, the problem reduces to a call option on the increment X_{T_1, T_2} :

$$\mathbb{E} \left[\left[X_{T_1, T_2} + (1 - K_2)F(0, T_2) + \left(1 - \frac{K_2}{B(T_1, T_2)} \right) x \right]^+ \middle| X_{T_1} = x \right] = C_{X_{T_1, T_2}}(k(x)),$$

where

$$C_{X_{T_1, T_2}}(k) = \mathbb{E}[(X_{T_1, T_2} - k)^+],$$

and

$$k(x) = (K_2 - 1)F(0, T_2) + \left(\frac{K_2}{B(T_1, T_2)} - 1 \right) x.$$

In the MA model, the increment X_{T_1, T_2} has an atom at

$$m = \mu_{T_2} - \mu_{T_1}$$

and exponential left and right tails. Therefore the call transform $C_{X_{T_1}, T_2}(k)$ is available in closed form. If A_L and A_R denote the left and right CDF coefficients of the MA increment, then

$$C_{X_{T_1}, T_2}(k) = \begin{cases} (m - k) - \frac{A_L}{p_{T_2}^-} \left(1 - e^{p_{T_2}^- (k-m)}\right) + \frac{A_R}{p_{T_2}^+}, & k < m, \\ \frac{A_R}{p_{T_2}^+} e^{-p_{T_2}^+ (k-m)}, & k \geq m. \end{cases}$$

The analytical time-zero price is then

$$V_0^{\text{MA}} = B(0, T_2) \int_{\mathbb{R}} C_{X_{T_1}, T_2}(k(x)) p_{X_{T_1}}(x) dx,$$

where $p_{X_{T_1}}$ is the asymmetric Laplace density of $X_{T_1} = f_{T_1, T_1}$. Since both $C_{X_{T_1}, T_2}(k(x))$ and $p_{X_{T_1}}(x)$ are piecewise exponential, the integral can be evaluated in elementary closed form. The real line is split at the location parameter μ_{T_1} of X_{T_1} and at the switching point

$$x^* = \frac{m - (K_2 - 1)F(0, T_2)}{\frac{K_2}{B(T_1, T_2)} - 1},$$

where $k(x) = m$. On each interval the integrand is a linear combination of exponentials and x -weighted exponentials, so the primitive is explicit.

This analytical price is used as a benchmark for the Monte Carlo estimator. The comparison validates both the simulation of the MA increments and the conditional CDF reconstruction used in the Lewis-FFT algorithm.

Table 10: MA model: comparison between analytical and Monte Carlo prices for the forward start call option with $T_1 = 6$ m, $T_2 = 1$ y, $K_2 = 1$ and $N_{\text{sim}} = 5\,000\,000$.

Analytical Price	MC Price
2.29685646	2.29424566

3.6 Minimal Additive Analytical CDF

In the Minimal Additive model, the CDF of the increment between two future dates can be computed in closed form. Let

$$X := f_{T_2} - f_{T_1}, \quad T_1 < T_2.$$

The CDF of the increment can be interpreted as the price of a digital payoff:

$$F_X(x) = \mathbb{P}(X \leq x) = \mathbb{E} \left[\mathbf{1}_{\{X \leq x\}} \right].$$

Using the Fourier representation of the indicator function, for a continuity point x one can write

$$F_X(x) = -\frac{1}{2\pi} \int_{\mathbb{R}+ia} e^{-iux} \frac{\varphi_X(u)}{iu} du, \quad 0 < a < p_2^-.$$

This is the analogue of the Lewis inversion formula for a digital payoff. Instead of the Fourier transform of a call payoff, the integrand contains the transform of the indicator function, which introduces the factor $1/u$.

In the MA model this integral can be evaluated analytically by the residue theorem. Indeed, after substituting the expression of φ_X , the integrand is a rational function multiplied by

$$e^{-iu(x-c)}.$$

Its poles are simple and are located at

$$u = ip_2^-, \quad u = -ip_2^+,$$

together with the pole at $u = 0$ generated by the digital payoff. For $x < c$, the contour is closed in the upper half-plane and the residue at $u = ip_2^-$ gives the left branch of the CDF. For $x > c$, the contour is closed in the lower half-plane and the residues at $u = -ip_2^+$ and $u = 0$ give the right branch.

The resulting CDF of the increment $X = f_{T_2} - f_{T_1}$ is

$$F_X(x) = \begin{cases} A_L e^{p_2^-(x-c)}, & x < c, \\ 1 - A_R, & x = c, \\ 1 - A_R e^{-p_2^+(x-c)}, & x > c, \end{cases}$$

where

$$A_L = \left(1 - \frac{p_2^-}{p_1^-}\right) \left[\frac{p_2^+}{p_1^+} + \left(1 - \frac{p_2^+}{p_1^+}\right) \frac{p_2^+}{p_2^- + p_2^+}\right],$$

and

$$A_R = \left(1 - \frac{p_2^+}{p_1^+}\right) \left[\frac{p_2^-}{p_1^-} + \left(1 - \frac{p_2^-}{p_1^-}\right) \frac{p_2^-}{p_2^- + p_2^+}\right].$$

At the atom location, the CDF is right-continuous. The left limit is

$$F_X(c^-) = A_L,$$

while

$$F_X(c) = 1 - A_R.$$

Hence, as discussed before, the distribution has an atom at c , with probability

$$\mathbb{P}(X = c) = F_X(c) - F_X(c^-) = 1 - A_L - A_R.$$

Using the expressions above, this mass is equal to

$$\mathbb{P}(X = c) = \frac{p_2^- p_2^+}{p_1^- p_1^+}.$$

Therefore, the MA increment distribution is mixed: it has a point mass at

$$c = \mu_2 - \mu_1$$

and an absolutely continuous component with exponential left and right tails.

Finally, since

$$f_{T_2} = f_{T_1} + X$$

and X is independent of f_{T_1} , the conditional CDF of f_{T_2} given $f_{T_1} = y$ is obtained by shifting the increment CDF:

$$\mathbb{P}(f_{T_2} \leq z \mid f_{T_1} = y) = F_X(z - y).$$

This analytical CDF is then used to simulate the MA increment by inverse transform, handling separately the two continuous branches and the atom at c .

3.7 MA Model: Comparison of Simulation Methods

Since the MA model admits a closed-form CDF for the increment, it is possible to compare the FFT-based simulation against an exact analytical benchmark. We consider three simulation methods for the increment X_{T_1, T_2} :

1. **FFT-Estimated:** the CDF is reconstructed via Lewis-FFT and the tail extrapolation uses the empirically estimated decay rates $\hat{\lambda}_{\pm}$;
2. **FFT-Theoretical:** the CDF is reconstructed via Lewis-FFT and the tail extrapolation uses the theoretical analyticity strip boundaries $p_t^{\pm}$;
3. **Analytic-CDF:** the increment is sampled directly by inverting the closed-form CDF, with no FFT involved.

All three methods use common random numbers for the Bernoulli draw and share the same seed, so that the comparison is not affected by Monte Carlo noise.

Table 11 reports the first four sample moments and the computational time for each method, together with the cost relative to the analytic reference.

Table 11: MA increment simulation: moments and timing comparison across three methods ($N_{\text{sim}} = 10^6$, seed = 2). Times are measured in seconds and refer to one representative MATLAB run on the same machine. *Cost vs Analytic* is the ratio of each method's runtime to that of the analytic CDF inversion.

Method	Time [s]	Cost vs Analytic	Mean	Variance	Skewness	Kurtosis
FFT-Estimated	0.1315	$1.7621\times$	0.003154	106.33	0.10167	15.363
FFT-Theoretical	0.1275	$1.7080\times$	0.003137	106.34	0.10037	15.389
Analytic-CDF	0.0746	$1.0000\times$	0.003139	106.34	0.10051	15.387

Table 12 reports the absolute differences in moments between the two FFT methods and the analytic reference.

Table 12: MA increment simulation: absolute difference in sample moments between FFT methods and the analytic CDF reference.

Moment	FFT-Estimated vs Analytic	FFT-Theoretical vs Analytic
Mean	1.4971×10^{-5}	2.7374×10^{-6}
Variance	5.0035×10^{-3}	7.1830×10^{-4}
Skewness	1.1678×10^{-3}	1.3244×10^{-4}
Kurtosis	2.4712×10^{-2}	2.1555×10^{-3}

The results confirm two conclusions. First, the FFT-Theoretical method is uniformly closer to the analytic reference than the FFT-Estimated method across all four moments, validating the choice of theoretical tail extrapolation discussed in Section 3.2. Second, the analytic CDF inversion is the fastest method, being about $1.7\times$ faster than the two FFT-based alternatives. This is expected, since the analytic method bypasses the FFT reconstruction entirely and inverts the closed-form CDF directly.. The remaining moment discrepancies between the FFT methods and the analytic reference are small and consistent with the numerical approximation introduced by the FFT-based CDF reconstruction.

4 Exotic Option Pricing

In this section we price the exotic products required in the assignment under the three calibrated linear additive models: Additive Bachelier (AB), Generalized Logistic (GL), and Minimal Additive (MA). The pricing is performed using the simulated increments between the two reset dates

$$T_1 = 6m, \quad T_2 = 1y.$$

In the linear additive framework, as already discussed, the terminal underlying is written as

$$S(T_2) = F(T_0, T_2) + X_{T_2}, \quad X_{T_2} = X_{T_1} + X_{T_1, T_2}.$$

Therefore,

$$S(T_2) - K_2 = X_{T_1} + X_{T_1, T_2} + F(T_0, T_2) - K_2.$$

According to the assignment, we set $K_2 = F(T_0, T_2)$, so that the moneyness adjustment $m_2 := F(T_0, T_2) - K_2$ is equal to zero. For the compound options we choose arbitrarily

$$K_1^{CoC} = K_1^{PoP} = 0.3.$$

and we maintain this definition also for the Risk Management section. The main numerical difficulty is that the value of each exotic payoff at T_1 depends on the conditional value of an inner vanilla option expiring at T_2 . A direct implementation would lead to a nested Monte Carlo procedure. To avoid the computational cost of a full nested simulation, we implemented two alternative estimators: a stochastic-mesh estimator based on interpolation and a no-grid estimator based on sorted inner increments.

4.1 Call-on-Call Option

The call-on-call option is a compound option whose underlying at T_1 is the value of a call option expiring at T_2 . For each realization $X_{T_1} = x$, the undiscounted inner call value is

$$c_{T_1}(x) = \mathbb{E} \left[(x + X_{T_1, T_2} + m_2)^+ \right],$$

where the expectation is computed with respect to the simulated distribution of the second increment X_{T_1, T_2} .

The call-on-call price is then

$$CoC = B(T_0, T_1) \mathbb{E}_{T_0} \left[(B(T_1, T_2) c_{T_1}(X_{T_1}) - K_1)^+ \right],$$

Using the **no-grid** implementation, the obtained prices are

$$CoC_{AB} = 5.5329, \quad CoC_{GL} = 5.5338, \quad CoC_{MA} = 5.5303.$$

The three models give similar values, with the GL model producing the largest call-on-call price. This is consistent with the fact that the CoC payoff is convex in the inner call value and is therefore sensitive to the right tail of the conditional distribution.

4.2 Put-on-Put Option

The put-on-put option is a compound option whose underlying at T_1 is the value of a put option expiring at T_2 . For each realization $X_{T_1} = x$, the undiscounted inner put value is

$$p_{T_1}(x) = \mathbb{E} \left[(-x - X_{T_1, T_2} - m_2)^+ \right].$$

The price of the put-on-put is

$$PoP = B(T_0, T_1) \mathbb{E}_{T_0} \left[(K_1 - B(T_1, T_2) p_{T_1}(X_{T_1}))^+ \right].$$

This payoff is positive when the discounted value of the inner put at T_1 is below the compound strike K_1 .

Using the **no-grid** implementation, the obtained prices are

$$PoP_{AB} = 0.0073, \quad PoP_{GL} = 0.0079, \quad PoP_{MA} = 0.0053.$$

The put-on-put prices are much smaller than the call-on-call and chooser prices. This happens because, with the selected value $K_1 = 0.3$, the event

$$B(T_1, T_2) p_{T_1}(X_{T_1}) < K_1$$

generates a relatively limited payoff in the simulated scenarios.

4.3 Chooser Option

The chooser option gives the holder, at T_1 , the right to choose whether the final payoff at T_2 will be a call or a put with strike K_2 . Its price is

$$Chooser = B(T_0, T_2) \mathbb{E}_{T_0} [\max(c_{T_1}(X_{T_1}), p_{T_1}(X_{T_1}))].$$

Unlike the compound options, the chooser option has no K_1 . Its value depends directly on the maximum between the conditional call and put values at the decision date.

Using the no-grid implementation, the obtained prices are

$$Chooser_{AB} = 10.38, \quad Chooser_{GL} = 10.38, \quad Chooser_{MA} = 10.378.$$

The chooser option is significantly more valuable than the PoP because it always selects the more valuable side between the conditional call and put. The values are close across the three models, with AB and GL essentially coinciding at the reported precision and slightly above MA.

4.4 Numerical Implementation and Interpolation Strategy

The pricing problem has the structure of a nested Monte Carlo, since for each outer scenario X_{T_1} one needs to compute the conditional value of a vanilla option expiring at T_2 . A naive nested Monte Carlo would require simulating a large number of inner paths for each outer path, leading to a computational cost of order

$$O(N_{\text{outer}}N_{\text{inner}}).$$

This is not convenient when N_{outer} is large. We therefore implemented two numerical strategies.

Stochastic-mesh approach. In the first approach, we build a grid on the simulated values of X_{T_1} . The grid is constructed using both quantiles and central equally spaced points. In particular, most grid points are assigned to empirical quantiles, while a smaller fraction is used to cover the central region with a regular grid. Tail quantiles and the minimum and maximum simulated values are also included to improve the stability of the interpolation.

For each grid point x_k , we compute

$$c_{T_1}(x_k) = \frac{1}{N} \sum_{j=1}^N \left(x_k + X_{T_1, T_2}^{(j)} + m_2 \right)^+,$$

and

$$p_{T_1}(x_k) = \frac{1}{N} \sum_{j=1}^N \left(-x_k - X_{T_1, T_2}^{(j)} - m_2 \right)^+.$$

The values $c_{T_1}(X_{T_1}^{(i)})$ and $p_{T_1}(X_{T_1}^{(i)})$ needed on the original Monte Carlo paths are then recovered through PCHIP interpolation. We choose PCHIP instead of cubic spline interpolation because PCHIP is shape-preserving and avoids artificial oscillations between grid points. This is particularly important in our case, since the conditional call and put values are smooth but monotone functions of X_{T_1} . A standard cubic spline may introduce spurious bumps, especially in regions where the grid is less dense, and these oscillations would have no financial interpretation in the conditional option value profile.

This method is much faster than a full nested Monte Carlo, but it still requires the evaluation of the inner expectation on all grid points. Moreover, it introduces an interpolation error, which depends on the number and placement of grid points.

No-grid approach. The second approach avoids both the stochastic mesh and the interpolation step. For each outer scenario we need to compute

$$c_i = \frac{1}{N} \sum_{j=1}^N \left(a_i + X_{T_1, T_2}^{(j)} \right)^+, \quad p_i = \frac{1}{N} \sum_{j=1}^N \left(-a_i - X_{T_1, T_2}^{(j)} \right)^+,$$

where

$$a_i = X_{T_1}^{(i)} + m_2.$$

Instead of looping over all inner increments for each a_i , the vector of increments $\Delta X_{T_1, T_2}$ is sorted once. Then cumulative counts and cumulative sums are used to compute the call and put expectations through the exercise threshold

$$X_{T_1, T_2} > -a_i.$$

This gives the same empirical inner expectations without constructing a grid on X_{T_1} . As a result, the method eliminates interpolation error and substantially reduces the computational time.

As a diagnostic check, the empirical inner put-call parity

$$c_i - p_i = a_i + \overline{\Delta X}_{T_1, T_2}$$

is verified. In the numerical test, the maximum absolute parity error is $9.3059 \cdot 10^{-9}$, which confirms the internal consistency of the no-grid implementation.

The timing benchmark is repeated for all the three calibrated models. This is useful because the computational comparison is not restricted to a single distributional assumption, but is tested on the same exotic products under AB, GL and MA dynamics.

Model	Grid time	No-grid time	Speed-up	Max abs. diff.	Max abs. diff. bps fwd
AB	44.527	2.3467	18.974x	$7.8357 \cdot 10^{-8}$	$2.0021 \cdot 10^{-5}$
GL	43.997	2.2196	19.822x	$6.6621 \cdot 10^{-8}$	$1.7023 \cdot 10^{-5}$
MA	45.322	1.8522	24.469x	$8.2655 \cdot 10^{-8}$	$2.1119 \cdot 10^{-5}$

Table 13: Comparison between stochastic-mesh pricing and no-grid pricing across the three calibrated models. Times are measured in seconds and refer to one representative MATLAB run on the same machine. The price difference is computed as no-grid price minus grid price.

The no-grid approach is consistently faster across all models. Although the absolute runtimes may vary across executions depending on the computational load of the machine, the speed-up remains of the same order of magnitude, around $20\times$. At the same time, the discrepancy between the grid and no-grid prices is negligible: the largest absolute price

difference is $8.2655 \cdot 10^{-8}$, corresponding to only $2.1119 \cdot 10^{-5}$ bps when measured with respect to the 12M forward. Therefore, the no-grid method removes the interpolation step and substantially reduces computational time without introducing a material pricing bias. For this reason, the no-grid prices are used as the reference prices in the model comparison.

4.5 Comparison Across Models and Bid-Ask Quotation

The final no-grid prices obtained under the three calibrated models are reported in Table 14. We use the no-grid prices as final prices because they are numerically indistinguishable from the grid-based ones, while being substantially faster from a computational point of view.

Product	AB	GL	MA
CoC	5.5329	5.5338	5.5303
PoP	0.0073	0.0079	0.0054
Chooser	10.3800	10.3800	10.3780

Table 14: No-grid Monte Carlo prices of the exotic options under the three calibrated linear additive models.

The three models produce very close prices for the call-on-call and chooser options. The largest price for the CoC is obtained under the GL model,

$$CoC_{GL} = 5.5338,$$

while the AB and GL prices for the chooser are essentially identical at the reported precision,

$$Chooser_{AB} \simeq Chooser_{GL} = 10.3800.$$

The put-on-put price is much smaller in absolute value and reaches its largest value under the GL model:

$$PoP_{GL} = 0.0079.$$

Overall, the model risk appears moderate in absolute price terms. The largest relative differences are observed for the PoP, but this is mainly due to the very low level of the option price.

4.5.1 Monte Carlo confidence intervals

The Monte Carlo standard errors are reported using two different normalizations. The first one divides the Monte Carlo standard error by the reference forward:

$$SE_{bps}^F = \frac{SE}{F(0, T_2)} 10^4.$$

The corresponding 95% confidence interval half-width is

$$CI_{bps}^{F,95\%} = \frac{1.96 SE}{F(0, T_2)} 10^4.$$

This is the relevant normalization for checking whether the Monte Carlo uncertainty is compatible with a bid-ask spread quoted in basis points of the underlying forward.

Product	Model	Price	MC SE	95% CI in price units	95% CI half-width, bps of forward
CoC	AB	5.5329	0.0033197	[5.5264, 5.5394]	1.6625
CoC	GL	5.5338	0.0032690	[5.5274, 5.5402]	1.6371
CoC	MA	5.5303	0.0034087	[5.5236, 5.5370]	1.7071
PoP	AB	0.0072807	0.0000160	[0.007249, 0.007312]	0.0080
PoP	GL	0.0079294	0.0000170	[0.007896, 0.007963]	0.0085
PoP	MA	0.0053465	0.0000138	[0.005319, 0.005374]	0.0069
Chooser	AB	10.3800	0.0035553	[10.3730, 10.3870]	1.7805
Chooser	GL	10.3800	0.0034735	[10.3732, 10.3868]	1.7395
Chooser	MA	10.3780	0.0038172	[10.3705, 10.3855]	1.9117

Table 15: Monte Carlo standard errors and 95% confidence intervals using the forward-based normalization.

All forward-based 95% confidence interval half-widths are below 5 bps of the forward. Therefore, the Monte Carlo uncertainty is sufficiently small to support a market-maker quote with a total bid-ask spread of 10 bps.

For completeness, we also report the same Monte Carlo uncertainty using the option price itself as denominator:

$$SE_{bps}^V = \frac{SE}{V_0} 10^4, \quad CI_{bps}^{V,95\%} = \frac{1.96 SE}{V_0} 10^4.$$

Product	Model	SE, bps of price	95% CI half-width, bps of price
CoC	AB	5.9999	11.760
CoC	GL	5.9074	11.578
CoC	MA	6.1637	12.081
PoP	AB	22.019	43.158
PoP	GL	21.404	41.952
PoP	MA	25.900	50.764
Chooser	AB	3.4252	6.7135
Chooser	GL	3.3462	6.5586
Chooser	MA	3.6780	7.2089

Table 16: Monte Carlo standard errors using the option-price-based normalization.

The price-based normalization is more restrictive, especially for the PoP. This is expected because the PoP has a very small price, so even a negligible absolute Monte Carlo error becomes large when divided by the option price itself. For this reason, we use the forward-based normalization to assess the 10 bps market-maker quote, while the price-based normalization is reported only as an additional relative-error diagnostic.

4.5.2 10 bps bid-ask quotation

The assignment requires the exotic products to be quoted with a maximum bid-ask spread of 10 bps. We interpret this as a bid-ask spread of 10 bps of the reference forward, centered around the no-grid Monte Carlo mid price. Therefore, the bid and ask prices are computed as

$$Bid = \max\left(0, V_0 - 5 \cdot 10^{-4} F(0, T_2)\right), \quad Ask = V_0 + 5 \cdot 10^{-4} F(0, T_2).$$

Using $F(0, T_2) \simeq 39.1369$, the half bid-ask spread is approximately

$$5 \cdot 10^{-4} F(0, T_2) \simeq 0.0196.$$

Product	Model	Mid price	Bid	Ask
CoC	AB	5.5329	5.5133	5.5525
CoC	GL	5.5338	5.5142	5.5534
CoC	MA	5.5303	5.5107	5.5499
PoP	AB	0.0072807	0.0000	0.02685
PoP	GL	0.0079294	0.0000	0.02750
PoP	MA	0.0053465	0.0000	0.02491
Chooser	AB	10.3800	10.3604	10.3996
Chooser	GL	10.3800	10.3604	10.3996
Chooser	MA	10.3780	10.3584	10.3976

Table 17: Market-maker quotation with a total bid-ask spread of 10 bps of the reference forward, corresponding to a 5 bps half-spread around the Monte Carlo mid price. For the PoP, the raw lower bound is negative and is therefore floored at zero, so the effective spread is below the 10 bps cap.

The forward-based Monte Carlo confidence intervals are narrower than the 10 bps bid-ask intervals. This means that the statistical uncertainty of the Monte Carlo estimator is smaller than the quotation tolerance required by the assignment. Therefore, the simulated prices are accurate enough to support a market-maker quote with a total bid-ask spread of 10 bps of the reference forward.

The only exception in terms of quotation construction is the PoP. Since its mid price is very close to zero, subtracting 5 bps of the reference forward would produce a negative bid. Since option prices are non-negative, we floor the bid at zero:

$$Bid = \max\left(0, V_0 - 5 \cdot 10^{-4} F(0, T_2)\right).$$

5 Closed-Form Formulas in the Minimal Additive Model

In this section we focus on the Minimal Additive model and derive elementary closed-form formulas for the Exotic options considered in the project. The aim is twofold. First, the analytical formulas provide a useful benchmark for the Monte Carlo pricing engine. Second, they exploit the specific structure of the MA model, whose conditional distribution is simple enough to make some nested expectations tractable.

We denote by Y the increment from t_0 to T_1 , and by Z the increment from T_1 to T_2 . In the MA model, the relevant quantities are

$$C_{MA} = \frac{\alpha_{MA}\beta_{MA}}{\alpha_{MA} + \beta_{MA}}, \quad \gamma_{MA} = \frac{1}{\alpha_{MA}} - \frac{1}{\beta_{MA}},$$

and

$$\mu_{T_i} = \gamma_{MA} \sigma_{T_i} \sqrt{T_i}, \quad i = 1, 2.$$

We also introduce

$$d = \mu_{T_2} - \mu_{T_1},$$

which is the location of the mass point of the conditional increment. In the derivation of the Call-on-Call and Put-on-Put formulas, we restrict the analysis to the region

$$-y < d,$$

where y is a realization of the first increment Y . This restriction is important because, in this region, the conditional payoff at T_1 has a simple analytical expression. As a consequence, the outer expectation

$$\mathbb{E}_0[(\cdot)^+]$$

can be reduced to a much simpler integral. The other regions can be treated analogously, but the objective of this section is to provide an analytical benchmark for the Monte Carlo engine. Therefore, we focus on this representative case, which is the one implemented in the numerical check.

The following notation is used:

$$p_{T_i}^+ = \frac{\beta_{MA}}{\sigma_{T_i} \sqrt{T_i}}, \quad p_{T_i}^- = \frac{\alpha_{MA}}{\sigma_{T_i} \sqrt{T_i}}, \quad i = 1, 2,$$

$$p_0 = \frac{p_{T_2}^- p_{T_2}^+}{p_{T_1}^- p_{T_1}^+},$$

$$A^- = p_0 \frac{(p_{T_1}^- - p_{T_2}^-)(p_{T_1}^+ + p_{T_2}^-)}{p_{T_2}^+ + p_{T_2}^-},$$

and

$$A^+ = p_0 \frac{(p_{T_1}^- + p_{T_2}^+)(p_{T_1}^+ - p_{T_2}^+)}{p_{T_2}^+ + p_{T_2}^-}.$$

Finally, we set

$$S = \sigma_{T_1} \sqrt{T_1}.$$

Before deriving the closed-form formulas, it is useful to recall the analytical structure of the Minimal Additive model. This is the key reason why the integrals appearing in the exotic option prices can be computed explicitly.

In the Minimal Additive model, the normalized random variable ζ follows an asymmetric Laplace distribution with density

$$p(\zeta) = C_{MA} \left[e^{\alpha_{MA}(\zeta - \gamma_{MA})} \mathbb{1}_{\{\zeta < \gamma_{MA}\}} + e^{-\beta_{MA}(\zeta - \gamma_{MA})} \mathbb{1}_{\{\zeta \geq \gamma_{MA}\}} \right],$$

The first increment can be written as

$$Y = S\zeta, \quad S = \sigma_{T_1} \sqrt{T_1}.$$

In the analytical derivation we do not need to explicitly introduce the density of Y . Instead, we keep the normalized MA density $p(\zeta)$ and perform the change of variables directly inside the integral:

$$y = S\zeta, \quad dy = S d\zeta.$$

Equivalently, for a generic function g ,

$$\mathbb{E}_0[g(Y)] = \int_{\mathbb{R}} g(y) f_Y(y) dy = \int_{\mathbb{R}} g(S\zeta) p(\zeta) d\zeta.$$

This is the change of variables used in the analytical computations. It is convenient because the density $p(\zeta)$ is piecewise exponential and therefore leads to elementary integrals.

The second ingredient is the distribution of the increment between T_1 and T_2 ,

$$Z := X_{T_2} - X_{T_1}.$$

In the MA model this increment has a mixed distribution: it has an atom and an absolutely continuous component. The atom is located at $d = \mu_{T_2} - \mu_{T_1}$, and its probability is $p_0 = \frac{p_{T_2}^- p_{T_2}^+}{p_{T_1}^- p_{T_1}^+}$. The continuous part of the increment distribution is

$$f_Z^{cont}(z) = A^- e^{p_{T_2}^-(z-d)} \mathbb{1}_{\{z < d\}} + A^+ e^{-p_{T_2}^+(z-d)} \mathbb{1}_{\{z > d\}},$$

Equivalently, the law of Z can be written as

$$dF_Z(z) = p_0 \delta_d(dz) + f_Z^{cont}(z) dz,$$

where δ_d denotes a Dirac mass at d .

This explicit representation is the core element behind the closed-form calculations. Once we condition on the first increment $Y = y$, the inner payoff only depends on the second increment Z . Hence, the conditional expectations reduce to integrals of linear payoff functions against exponential densities, plus the contribution of the atom at d .

For the Call-on-Call and Put-on-Put formulas, we focus on the region

$$-y < d.$$

Since $y = S\zeta$, this condition becomes

$$\zeta > -\frac{d}{S}.$$

Therefore, the outer expectation can be written directly with respect to the normalized MA density:

$$\mathbb{E}_0[(\cdot)^+] = \int_{-d/S}^{+\infty} (\cdot) p(\zeta) d\zeta.$$

In the region $-y < d$, the inner conditional Call and Put values have elementary expressions. The resulting outer integrals are therefore one-dimensional integrals with respect to ζ , where the integrand is a linear/exponential function multiplied by a piecewise exponential density. This is the practical reason why the closed-form formulas can be obtained in the MA model.

5.1 Analytical Formula for the Call-on-Call

The Call-on-Call payoff is

$$CoC(t_0; T_1, K_1; T_2, K_2) = B(t_0, T_1) \mathbb{E}_0 \left[\left(B(T_1, T_2) \mathbb{E}_1[(S_{T_2} - K_2)^+] - K_1 \right)^+ \right].$$

In the project, the second strike is given as

$$K_2 = F(t_0, T_2),$$

so that the inner option is written at-the-money with respect to the initial T_2 -forward.

Conditionally on $Y = y$, the inner call price is an increasing function of y . This monotonicity is crucial. It allows us to choose K_1 as the value that makes the outer payoff switch exactly at the boundary of the simplified region. In particular, in the case

$$-y < d,$$

we define

$$E_1^C(y) = A^- \left[\frac{d+y}{p_{T_2}^-} - \frac{1}{(p_{T_2}^-)^2} + \frac{1}{(p_{T_2}^-)^2} e^{-p_{T_2}^-(y+d)} \right] + A^+ \left[\frac{d+y}{p_{T_2}^+} + \frac{1}{(p_{T_2}^+)^2} \right] + p_0(d+y).$$

Then K_1 is selected as

$$K_1^{CoC} = B(T_1, T_2) E_1^C(-d).$$

The choice of K_1 is made so that the exercise region of the outer option coincides with $y > -d$. This is the same condition implemented in the code: since the inner call price is increasing in y , choosing K_1 at the boundary $y = -d$ simplifies the positivity condition

in the outer payoff. The analytical integration is then performed only over the relevant exponential tail of the MA density.

The resulting analytical price is

$$\begin{aligned} CoC_{analytic}^{MA} = & C_{MA}B(t_0, T_2) \left[\frac{A^+}{p_{T_2}^+ \beta_{MA}} e^{\beta_{MA}(\gamma_{MA} + \frac{d}{S})} \left(\frac{S}{\beta_{MA}} + \frac{1}{p_{T_2}^+} \right) + \right. \\ & + \frac{A^-}{p_{T_2}^-} e^{\beta_{MA}(\gamma_{MA} + \frac{d}{S})} \left(\frac{S}{\beta_{MA}^2} - \frac{1}{p_{T_2}^- \beta_{MA}} + \frac{1}{p_{T_2}^- (Sp_{T_2}^- + \beta_{MA})} \right) + \\ & \left. - \frac{K_1^{CoC}}{\beta_{MA}B(T_1, T_2)} e^{\beta_{MA}(\gamma_{MA} + \frac{d}{S})} + p_0 \frac{S}{\beta_{MA}^2} e^{\beta_{MA}(\gamma_{MA} + \frac{d}{S})} \right]. \end{aligned}$$

5.2 Analytical Formula for the Put-on-Put

The Put-on-Put is treated in the same spirit. The payoff is

$$PoP(t_0; T_1, K_1; T_2, K_2) = B(t_0, T_1) \mathbb{E}_0 \left[\left(K_1 - B(T_1, T_2) \mathbb{E}_1[(K_2 - S_{T_2})^+] \right)^+ \right].$$

Again, the key simplification is obtained in the region $-y < d$.

In this region, the conditional put expectation has the simple form

$$E_1^P(y) = \frac{A^-}{(p_{T_2}^-)^2} e^{-p_{T_2}^-(y+d)}.$$

The first strike is then selected as

$$K_1^{PoP} = B(T_1, T_2) E_1^P(-d).$$

This choice makes the positive part in the outer expectation analytically tractable, exactly as in the Call-on-Call case.

The analytical Put-on-Put price implemented in the code is

$$PoP_{analytic}^{MA} = C_{MA}B(t_0, T_2) e^{\beta_{MA}(\gamma_{MA} + \frac{d}{S})} \left[\frac{K_1^{PoP}}{\beta_{MA}B(T_1, T_2)} - \frac{A^-}{(p_{T_2}^-)^2 (Sp_{T_2}^- + \beta_{MA})} e^{\beta_{MA}(\gamma_{MA} + \frac{d}{S})} \right].$$

The formula is more compact than the Call-on-Call one because, in the considered region, the conditional put expectation contains only one exponential term.

5.3 Analytical Formula for the Chooser Option

The Chooser option is

$$Chooser(t_0; T_1; T_2, K_2) = B(t_0, T_2) \mathbb{E}_0 \left[\max \left(\mathbb{E}_1[(S_{T_2} - K_2)^+], \mathbb{E}_1[(K_2 - S_{T_2})^+] \right) \right].$$

In the project, we again take $K_2 = F(t_0, T_2)$. Differently from the CoC and PoP cases, the Chooser does not require the introduction of a first strike K_1 . The payoff is determined by comparing the conditional value of a call and the conditional value of a put at time T_1 .

In the MA model, the analytical expression used in the code is

$$Chooser_{analytic}^{MA} = \frac{C_{MA}}{\beta_{MA}^2} e^{\beta_{MA} \gamma_{MA}} \left[B(t_0, T_2) \sigma_{T_2} \sqrt{T_2} + B(t_0, T_1) \sigma_{T_1} \sqrt{T_1} \right].$$

This formula is particularly simple and shows that, in the MA model, the Chooser price depends on the two volatility scales

$$\sigma_{T_1} \sqrt{T_1}, \quad \sigma_{T_2} \sqrt{T_2},$$

weighted by the corresponding discount factors.

Remark: the analytical Chooser formula is obtained by repeatedly using put-call parity and the closed-form expression for European call options in the Minimal Additive model. In our calibrated setting, the derivation is simplified by the fact that the option is written with $K_2 = F(t_0, T_2)$ and by the negative sign of γ_{MA} , which determines the relevant branch of the MA closed-form formula.

5.4 Comparison with Monte Carlo Prices

The numerical prices are obtained using the same nested structure as in the Exotic pricing section, but with a computational shortcut. A full nested Monte Carlo would require simulating a large number of inner paths for each outer path. With $5 \cdot 10^6$ outer simulations and $5 \cdot 10^6$ inner simulations, this would be computationally infeasible. Therefore, the conditional value at T_1 is computed on a smaller grid of y -values and then interpolated on the simulated realizations of the first increment.

In the current implementation, the conditional expectations are computed on a grid of 2000 points, and then interpolated using the `pchip` method. This avoids oscillations and gives a stable approximation of the conditional pricing function. It is important to stress that this analytical check is performed under the tractable strike choice described above, not under the main exotic-pricing choice $K_1 = 0.3$. Therefore, the prices in this table are not meant to coincide with the MA prices reported in Section 14; they are used only to validate the numerical implementation against a closed-form benchmark.

Product	Monte Carlo price	Analytical price	Absolute difference
Call-on-Call	4.0264	4.0292	0.0028
Put-on-Put	0.4903	0.5088	0.0185
Chooser	10.3788	10.3807	0.0019

Table 18: Comparison between Monte Carlo and analytical prices in the Minimal Additive model.

The agreement between Monte Carlo and analytical prices is satisfactory. The Call-on-Call and the Chooser prices are very close, while the Put-on-Put shows a slightly larger discrepancy. This is still reasonable, given the numerical approximations involved in the Monte Carlo pricing engine.

The difference between numerical and analytical prices can be mainly attributed to two sources.

First, the numerical method avoids a fully nested Monte Carlo as discussed above. Instead of using $5 \cdot 10^6 \times 5 \cdot 10^6$ simulations, the conditional payoff is priced only on a grid and then interpolated. This approximation is much faster, but it necessarily introduces an interpolation error.

Second, the simulation of the increments is based on the CDF obtained through FFT inversion. In the tails, the numerical CDF may not cover the whole interval $[0, 1]$. For instance, if the computed CDF reaches only 0.9996 and a uniform draw is equal to 0.9997, the inversion requires extrapolation beyond the reliable FFT range. This is delicate and may generate pricing errors, especially for payoffs that are sensitive to tail events.

Overall, the analytical formulas confirm the reliability of the Monte Carlo pricing engine in the MA model. At the same time, the comparison highlights the two main numerical limitations of the simulation-based approach: interpolation in the conditional pricing step and tail management in the FFT-based CDF inversion.

6 Risk Management

The objective of the Risk Management part of the Project is to hedge the portfolio of Exotic options priced in the previous section by using liquid Plain Vanilla options and the underlying front Future. The hedging strategy is designed as a first-order hedge with respect to the main risk factors of the portfolio: the forward level and the term structure of normal volatility.

The portfolio considered in this section is composed of short positions in the three

exotic products:

$$\Pi_0^{\text{exo}} = -V_0^{\text{CoC}} - V_0^{\text{PoP}} - V_0^{\text{Chooser}}.$$

The risk management engine is based on the Additive Bachelier model. This choice is motivated by its overall calibration quality: AB provides the lowest global RMSE and one of the best fits around the maturities relevant for the hedging exercise. Although GL is marginally better at the 12M maturity, the differences are very small, and AB remains a robust benchmark model for the risk-management implementation. **Remark:** the same analysis can be conducted for a different portfolio composition, for instance by considering only the CoC and PoP, or by changing the number of short positions in each exotic product. This only requires modifying the portfolio configuration in the *Risk_Management.m* script. We interpret this portfolio from the perspective of a market maker who has sold the three exotic products and wants to hedge the resulting risk exposure.

6.1 Hedging instruments

The assignment requires the exotic portfolio to be hedged using Plain Vanilla options and the underlying front Future. It does not prescribe which Plain Vanilla options must be selected. Therefore, the choice of the hedging instruments is itself part of the quantitative hedging rule. In the following paragraphs we present and discuss the Quantitative Rule that we have followed to hedge our portfolio:

The portfolio contains exotic products whose payoff structure depends on the two key dates T_1 and T_2 , corresponding respectively to the 6-month and 12-month horizons. For this reason, the Plain Vanilla options used for hedging are selected on the same two maturity buckets:

- Plain Vanilla call with maturity close to T_1 , i.e., close to 6 months, namely the option with maturity 17th November 2020;
- Plain Vanilla call with maturity close to T_2 , i.e., close to 12 months, namely the option with maturity 17th May 2021;
- The front future, corresponding to the first forward maturity strictly later than the valuation date, namely 17th August 2020.

This choice is consistent with the risk profile of the Exotic portfolio. The Exotics are mainly exposed to movements of the forward level and to changes in the volatility term structure at the two relevant horizons. Hence, using Plain Vanilla options with maturities close to T_1 and T_2 allows us to hedge the two main volatility pillars driving the portfolio value.

Within each selected maturity bucket, the strike is chosen according to an ATM criterion. More precisely, for the 6-month maturity we choose the call option whose strike is closest to the corresponding market-implied forward:

$$K_{\text{ATM}}^{6M} = \arg \min_{K \in \mathcal{K}_{6M}^{\text{Call}}} |K - F(0, T_1)|,$$

while for the 12-month maturity we choose

$$K_{\text{ATM}}^{12M} = \arg \min_{K \in \mathcal{K}_{12M}^{\text{Call}}} |K - F(0, T_2)|.$$

The ATM criterion is natural because ATM options are usually the most liquid contracts and provide the largest Vega exposure. Therefore, they are the most effective instruments to hedge the volatility risk of the Exotic portfolio.

The decision to use Calls rather than puts is also part of the quantitative rule. We decided to use Calls rather than Puts for two main reasons. The first one is because in the available dataset, Call options provide a richer set of quoted strikes around the relevant maturities than Put options. For this reason, calls are preferred, since they allow a more robust and reproducible ATM selection. The second reason is given directly by the following table:

Side	N	RMSE AB	MAE AB	Mean Error AB
Call	409	0.090016	0.069879	-0.007078
Put	187	0.120810	0.079995	-0.022503

Table 19: Calibration quality by side

The table shows that the AB calibration errors are lower on the call side than on the put side. This supports the choice of calls as hedging instruments in our implementation. This kind of analysis leads us to use the Calls, but it is important to underline that using puts with the same maturity and moneyness would lead to comparable exposures, up to put-call parity effects.

The front Future is used only to hedge the residual delta exposure after the two Plain Vanilla options have been selected. Since a Futures contract has no direct volatility exposure, its Vega is set equal to zero:

$$\nu_F^{6M} = 0, \quad \nu_F^{12M} = 0, \quad \Delta_F = 1.$$

6.2 Finite-difference Greeks

The hedge is based on **central finite-difference** estimates of Delta and Vega. The bumps are **absolute bumps**, because both the Forward Price and the Bachelier Normal Volatility are expressed in price units:

$$\varepsilon_F = 0.01, \quad \varepsilon_\sigma = 0.01.$$

For each instrument i , the Delta is computed as

$$\Delta_i = \frac{V_i(F + \varepsilon_F) - V_i(F - \varepsilon_F)}{2\varepsilon_F}.$$

For the exotic products, the bumped forward is the 12-month forward $F(0, T_2)$, while the contractual strike $K_2 = F(0, T_2)$ is kept fixed. This is important: after the trade date, K_2 is a contractual quantity and must not move with the bumped market forward.

The Vega is computed separately with respect to the two relevant volatility pillars:

$$\nu_i^{6M} = \frac{V_i(\sigma_{6M} + \varepsilon_\sigma, \sigma_{12M}) - V_i(\sigma_{6M} - \varepsilon_\sigma, \sigma_{12M})}{2\varepsilon_\sigma},$$

and

$$\nu_i^{12M} = \frac{V_i(\sigma_{6M}, \sigma_{12M} + \varepsilon_\sigma) - V_i(\sigma_{6M}, \sigma_{12M} - \varepsilon_\sigma)}{2\varepsilon_\sigma}.$$

For the 6-month Plain Vanilla option, the 12-month Vega is set equal to zero, since the contract ceases to exist before the shock at 12-month:

$$\nu_{PV,6M}^{12M} = 0.$$

All sensitivities are computed using common random numbers. In practice, the same uniform random variables are reused in the up and down scenarios. This reduces the Monte Carlo noise in the finite-difference estimator and makes the computed Greeks more stable.

The portfolio Greeks before hedging are given by the sum of the Greeks of the three short exotic positions:

$$\Delta_\Pi = -\Delta_{CoC} - \Delta_{PoP} - \Delta_{Chooser},$$

$$\nu_\Pi^{6M} = -\nu_{CoC}^{6M} - \nu_{PoP}^{6M} - \nu_{Chooser}^{6M},$$

$$\nu_\Pi^{12M} = -\nu_{CoC}^{12M} - \nu_{PoP}^{12M} - \nu_{Chooser}^{12M}.$$

6.3 Quantitative hedging rule

The quantitative hedging rule is composed of two layers.

The first layer concerns the systematic selection of the hedging instruments. Since the Exotic portfolio depends on the two key dates T_1 and T_2 , we use Plain Vanilla options with maturities close to 6 months and 12 months. Within each maturity bucket, we select the Call option whose strike is closest to the corresponding market-implied forward. We choose to use calls instead of puts as discussed at the end of Section 6.1.

The second layer concerns the determination of the hedge quantities. Once the hedging instruments have been selected, the hedge is built through a **cascade rule**. The idea is to start from the longest volatility exposure and then move backwards to the shorter one. This produces a triangular hedging system: the 12-month Call is used first to hedge the 12-month Vega, the 6-month Call is then used to hedge the remaining 6-month Vega, and the front Future is finally used to hedge the residual delta.

Operationally, the 12-month Vega is hedged using the 12-month ATM Plain Vanilla Call:

$$\tilde{q}_{12M} = -\frac{\nu_{\Pi}^{12M}}{\nu_{PV,12M}^{12M}}.$$

Then, the 6-month Vega is hedged using the 6-month ATM Plain Vanilla Call, taking into account the 6-month Vega generated by the already selected 12-month Call:

$$\tilde{q}_{6M} = -\frac{\nu_{\Pi}^{6M} + q_{12M}\nu_{PV,12M}^{6M}}{\nu_{PV,6M}^{6M}}.$$

Finally, after adding the Delta contribution of the two hedging options, the remaining Delta is hedged with the front Future:

$$\tilde{q}_F = -(\Delta_{\Pi} + q_{6M}\Delta_{PV,6M} + q_{12M}\Delta_{PV,12M}).$$

Following the technique implemented in practice, we decided that the hedge must be implemented with integer quantities, the continuous hedge ratios are rounded towards zero:

$$q_j = \text{trunc}_0(\tilde{q}_j),$$

where

$$\text{trunc}_0(x) = \begin{cases} \lfloor x \rfloor, & x \geq 0, \\ \lceil x \rceil, & x < 0. \end{cases}$$

This rule is deliberately conservative. It avoids over-hedging and limits transaction costs, at the price of **leaving a small residual exposure**. A nearest-integer rule would produce a more aggressive hedge, but could also increase the hedging cost and the risk of over-adjusting the portfolio.

Important: The truncation is applied sequentially: after computing $\tilde{q}_{12M}$, we obtain q_{12M} and use this integer quantity in the computation of $\tilde{q}_{6M}$. The same logic is then applied to the front Future.

The resulting hedge quantities are:

$$q_{6M} = 1, \quad q_{12M} = 1, \quad q_F = 0.$$

The zero position in the front Future means that, after adding the two ATM Calls, the remaining delta exposure is sufficiently small to be rounded to zero by the chosen integer rule.

6.4 Hedging cost

The assignment specifies bid-ask hedging costs of 1 bp for the underlying and 4 bp for each Plain Vanilla option. We interpret these quantities as full bid-ask spreads. Since the hedging portfolio is set up from mid prices, the one-way transaction cost is therefore equal to one half of the quoted bid-ask spread.

$$C_{\text{hedge}} = \frac{1}{2} \cdot 10^{-4} |q_F| + \frac{1}{2} \cdot 4 \cdot 10^{-4} (|q_{6M}| + |q_{12M}|).$$

Using the hedge quantities above, the total hedging cost is

$$C_{\text{hedge}} = 0.0004.$$

6.5 Initial cash-flow

At inception, the position generates an initial cash-flow due to the sale of the Exotic portfolio and to the purchase of the hedging Plain Vanilla options. We use the convention that a positive cash-flow corresponds to cash received at inception. Since the Exotic portfolio is short, the premiums of the three Exotic products are received, while the premiums of the hedging Plain Vanilla options are paid. The front Future does not enter the initial cash-flow, since a Futures contract has zero initial premium under the standard marking-to-market convention. Its effect appears only through the subsequent variation of the front Future price.

The initial cash-flow is therefore computed as

$$CF_0 = - \left(\sum_{i \in \{\text{CoC}, \text{PoP}, \text{Chooser}\}} n_i V_i(0) + q_{6M} C_{6M}(0) + q_{12M} C_{12M}(0) \right) - C_{\text{hedge}},$$

where $n_i = -1$ for each Exotic product. With the hedge quantities selected by the integer cascade rule, the initial net cash-flow is

$$CF_0 = 6.2996.$$

This positive value means that the premium received by selling the Exotic products is larger than the cost of setting up the hedging portfolio. In the backtesting exercise we do not include the cash account in the reported P&L. The reported P&L measures the mark-to-market change of the portfolio between 2 June 2020 and the two following Tuesdays. Equivalently, we assume that the initial cash-flow is kept in a cash account and that its accrual over one or two weeks is negligible.

6.6 Backtesting on the following Tuesdays

The hedge is tested on the next two Tuesdays, namely on the 9th June 2020 and 16th June 2020. On each date, the market curves and the volatility surface are rebuilt from the available option data. The Exotic products are repriced under the Additive Bachelier model using the remaining time to the original reset dates. The hedging calls are valued using the market prices observed at the same strikes chosen on the initial date. This approach introduces an availability risk, because it assumes that the same hedging option strikes are quoted also on the two backtesting dates. In our case the prices of the Calls related to the strikes were both available on the next two Tuesdays. If they were not available, we would probably have chosen to continue with another strategy: get Calls' prices with a strike further than the ATM one, but with prices available in each of the three dates that we needed.

The hedged P&L is computed as

$$\begin{aligned} \text{P\&L}^H(t) = & \sum_{i \in \{\text{CoC}, \text{PoP}, \text{Chooser}\}} n_i (V_i(t) - V_i(0)) + q_{6M} (C_{6M}(t) - C_{6M}(0)) \\ & + q_{12M} (C_{12M}(t) - C_{12M}(0)) + q_F (F_{\text{front}}(t) - F_{\text{front}}(0)) - C_{\text{hedge}}, \end{aligned}$$

where in our case $n_i = -1$ for each exotic product. The unhedged P&L is obtained by considering only the first term, namely the change in value of the Exotic portfolio.

Date	Unhedged P&L	Hedged P&L	Hedge Improvement
9 June 2020	-1.401	0.7587	2.1596
16 June 2020	-2.5577	0.1019	2.6596

Table 20: Comparison between unhedged and hedged P&L on the two backtesting dates.

The hedge improves the stability of the portfolio. On the first Tuesday, the unhedged Exotic portfolio generates a loss of -1.401 , while the hedged portfolio produces a P&L of 0.7587 . The hedge not only offsets the loss, but leads to a positive P&L after transaction costs. This should not be interpreted as an arbitrage profit, but as the result of a first-order hedge combined with the realized market movement over the week.

On the second Tuesday, the unhedged portfolio loss is larger, -2.5577 . After hedging, the portfolio produces a positive P&L of 0.1019 . The reduction in absolute P&L is therefore stronger on the second backtesting date. This suggests that the selected ATM Calls capture a large part of the volatility exposure of the Exotic portfolio.

The hedge is not perfect for several reasons. First, the strategy only hedges first-order Delta and Vega exposures. It does not explicitly hedge Gamma, Volga, Vanna, or other higher-order effects, which can be relevant for compound and Chooser options. Second, the hedge quantities are constrained to be integer positions as discussed above. The theoretical hedge ratios are continuous quantities: for example, a perfect first-order hedge could require buying 1.1456 Plain Vanilla Calls, while in practice the implemented position must be rounded to 1 Calls according to the chosen truncation rule. This necessarily leaves a residual unhedged exposure. Third, the front Future is not exactly the same risk factor as the 12-month forward driving the exotic payoff, so a basis risk remains. Finally, the Exotic products are repriced with a model, while the hedging options are marked using market prices; this introduces model-to-market basis risk.

Overall, the Quantitative Rule is simple, reproducible and consistent with the instruments allowed by the project. The cascade structure is particularly useful because it first neutralizes the long-maturity Vega, then the short-maturity Vega, and only at the end the residual Delta. The empirical backtest on the two following Tuesdays shows that the hedge materially reduces the losses of the Exotic portfolio on both backtesting dates, even after including transaction costs.

7 Conclusions

This project has implemented the full workflow of exotic derivative pricing and risk management under three linear additive models, Additive Bachelier (AB), Generalized Logistic (GL), and Minimal Additive (MA), applied to WTI crude oil options during the Covid-19 period of June 2020.

Calibration. The cascade calibration scheme extends naturally to the MA and GL models whenever the shape parameters are time-independent. The AB and GL models achieve comparable global RMSE (0.101 and 0.105), outperforming the MA model (0.134), particularly in the near-ATM region. This is consistent with the MA model's asymmetric Laplace density, which transitions abruptly from its central cusp to the exponential tail regime without a smooth intermediate zone. A structural identifiability issue was found: only the ratio α/β is robustly determined from OTM option prices, while the individual parameters depend on the initial guess.

Monte Carlo simulation. The Lewis-FFT-S algorithm provides accurate and

efficient CDF inversion for all three models. A key finding is that theoretical tail extrapolation using the analyticity strip boundaries $p_t^\pm$ is significantly more reliable than empirically estimated rates. The MA model requires a Bernoulli decomposition to handle the probability atom in the increment distribution; the analytical CDF inversion available in the MA model confirms the accuracy of the FFT-based approach and represents the fastest simulation method.

Exotic pricing. The no-grid method outperforms the stochastic-mesh approach by roughly $20\times$ in computational time while introducing negligible pricing error (below 10^{-7}). The three models produce close exotic prices: model risk is moderate, with the largest spread observed for the call-on-call, the most sensitive product to right-tail behaviour. Closed-form MA benchmark formulas, derived in tractable cases, confirm the reliability of the Monte Carlo engine.

Risk management. The first-order hedge using two ATM plain vanilla calls materially reduces the portfolio losses on both backtesting dates, turning losses of -1.40 and -2.56 into positive P&Ls of $+0.76$ and $+0.10$. Residual hedge error is attributable to unhedged higher-order Greeks, integer rounding of hedge quantities, and model-to-market basis risk.

Overall. Linear additive models provide a coherent and tractable framework for the full pricing chain. Their main limitation is the restricted shape flexibility of the MA model near ATM and the common difficulty in fitting long-maturity options, where market liquidity is lower. Extensions to time-dependent shape parameters represent a natural direction for future work.

References

- [1] Azzone, M., & Baviera, R. (2023). A fast Monte Carlo scheme for additive processes and option pricing. *Computational Management Science*, 20(1), 31.
- [2] Azzone, M., & Baviera, R. (2021). Synthetic forwards and cost of funding in the equity derivative market. *Finance Research Letters*, 41, 101841.
- [3] Baviera, R., & Massaria, M.D. (2026). The Additive Bachelier model with an application to the oil option market in the Covid period, *Journal of Computational and Applied Mathematics*, 487, 117741.
- [4] Carr, P., & Torricelli, L. (2021). Additive logistic processes in option pricing. *Finance and Stochastics*, 25(4), 689-724.
- [5] Baviera, R., & Manzoni, P. (2026). Fast and General Simulation of Lévy-driven OU processes for Energy Derivatives.
- [6] Baviera, R., (2026), The instantaneous forward.